

Lucid Dawn: Dream Survivor — Full UI/UX Spec (out-game + in-game)

Annotated-wireframe spec. Each screen has ① the real game UI it references (web-harvested with `ui-ref`, embedded inline), ② a wireframe derived from it (`wireframes/*.svg`), ③ a numbered legend (every element boxed/circled with a leader line pulled outside the frame), ④ state matrix · input parity · data binding, ⑤ a plain-language **UX rationale**. Editable source = this `.md` ; shareables = wide 16:9 PDF/DOCX (Appendix C). Companion docs: **design tokens** `lucid_dawn_ui_ux_tokens.md` , **decision/number tracker** `lucid_dawn_ui_ux_decisions.md` (shared by the EN/中文/한글 versions).

Reference sources. Images were harvested with **Game UI Reference CLI (`ui-ref`)** from **interfaceingame.com** per-game pages (Hades · Risk of Rain 2 · Honkai: Star Rail · Slay the Spire · Returnal · Hollow Knight · Moonlighter · Destiny 2). Personal research citations, not a redistributable asset pack. Index/recipe: Appendix B. Direct survivor-like patterns (Vampire Survivors/Brotato/20MTD) are described in words in §B-2 (not on the harvest site).

0. Cover

field	value
Document	Lucid Dawn: Dream Survivor — Full UI/UX Spec (out-game + in-game)
Game / build	Lucid Dawn: Dream Survivor / Vertical Slice (v0.8 scope, v0.9 GDD)
Version / date / author	v1.0 (EN master) / 2026-06-30 / UX
Status	draft
Source GDD	<code>GameDesign/lucid_dawn_design_docs/lucid_dawn_gdd.md</code> (§2 loop, §4 systems, §5 chars, §6 stage, §7 boss, §9 UI/UX, §10 BM/records, §12 TDD handoff)
Wireframes	out-game <code>title-mainmenu/character-select/stage-select/settings/meta-progress/leaderboard.svg</code> ; in-game <code>hud/hud-boss/levelup/skilltree/results/pause.svg</code> ; <code>flow.svg</code>

1. Overview & goals

- **Problem this solves:** GDD §9-1 (screen flow) and §9-2 (HUD table) only fix "what goes where". To build it you need every screen's states, inputs, data binding, edge cases, accessibility, and the *why* — from the menus (out-game) through combat/reward/result (in-game).
- **Core experience (GDD §1-1):** "pressure → split-second read → perfect dodge → lucid rush → purify/grow → waking up by your own hand". The UI must never hide that loop.
- **Measurable success**
 - Out-game: a first-time player starts a run within **3 minutes** (menu depth + label clarity).
 - In-game: at any instant the player reads **time left until 7:00** within 1s (§3-2); at the 180-enemy hard cap (§6-4) the player, lethal bullets, hazard fields and pickups stay distinguishable by **color + shape** (§13-3 #5).

- Reward flow follows GDD §12-3: **skill point grant** → **skill tree** → **separate 4-item pick**.
- BM (§10): no paid power, no gacha → the meta screen is **play-unlock only (no purchase UI)**.
- **In scope**: out-game O1 Title/Main, O2 Character select, O3 Stage/Difficulty, O4 Options, O5 Meta progress (unlock/codex), O6 Leaderboard; in-game I1 HUD, I2 HUD (boss/hidden), I3 Level-up/item pick, I4 Boss reward (tree + 4-pick), I5 Pause, I6 Result (3 branches).
- **Out of scope (this doc)**: co-op-only UI (lobby/team gauge/ghost/joint ult — GDD §8) is unbuilt in VS → tracked as risk in §6-1; tutorial screens, shop (none per BM), cutscene polish are later.

2. Users & context

- **Persona**: survivor-like / 20-min-survival / bullet-hell / action-roguelite / co-op-PvE players (GDD §1). Novice (first Dawn Wake) through veteran (Dream Break / no-damage records) coexist.
- **Platform / input**: PC/Steam first. Keyboard+mouse and gamepad both (GDD §2-5). Touch out of scope (design stays pad-parity-extensible).
- **Entry context**: out-game = non-combat, focus-driven; in-game = real-time combat, except level-up/reward picked under pause/protection (§4-5, §7-1). One run = 6:40→7:00 (20 min, internal 0–1200s). HUD only in-run; hidden in menus/result. VS roster = Kohaku, Toko (4 others locked); stage = LD-001 (§3, §5, §6-2).

2-1. Design principles (every screen)

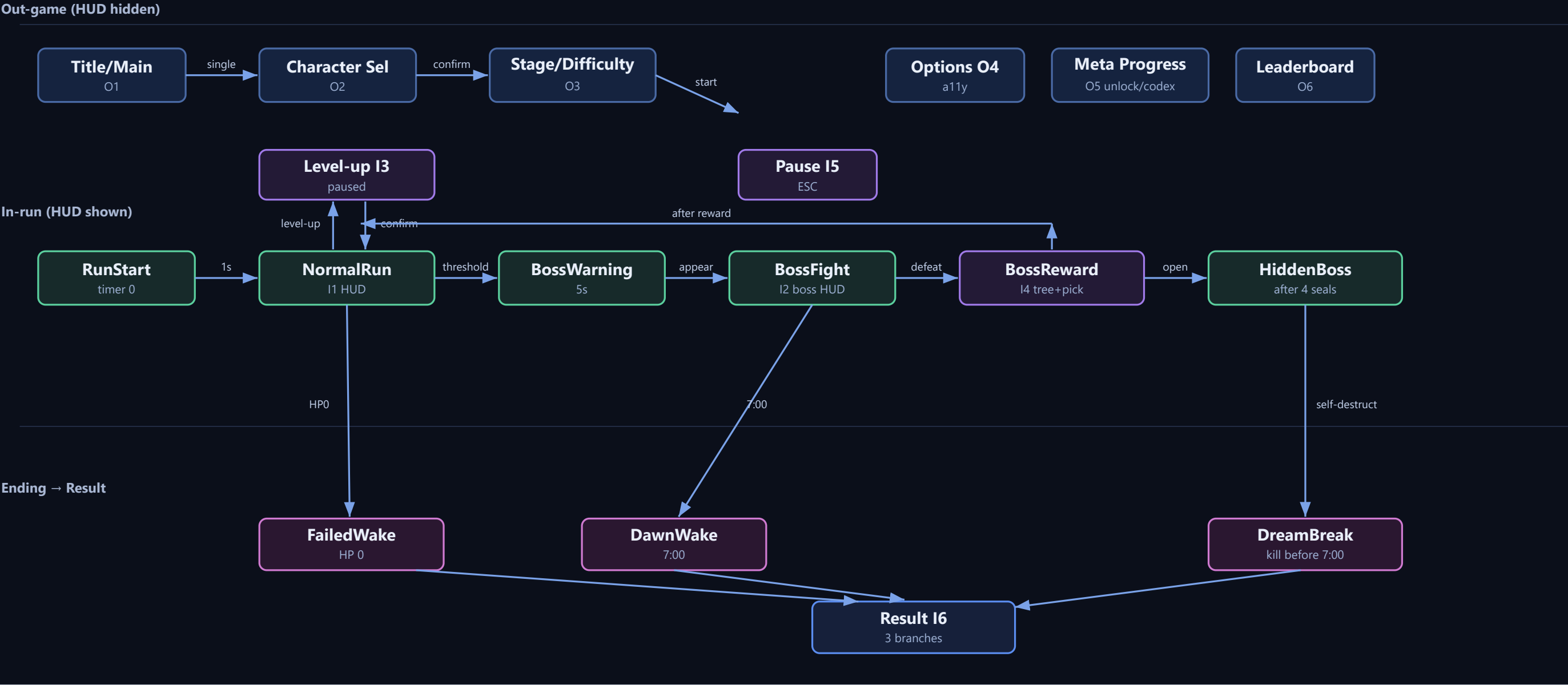
#	principle	meaning here	if broken
P1	instant feedback	every action/state change answers at once (sound + display)	weak feel/awareness
P2	hazard legibility	player > immediate threats (bullets/zones/charges) > pickups > ambient	danger unseen
P3	legible in chaos	at 180 enemies the core stays color+shape+sound; cap juice	one-shot patterns buried
P4	consistency	lock position/color/icon/control/bar shape across screens; flag deliberate exceptions	constant confusion
P5	progressive disclosure	core first, rest when needed (lucid/purify/boss taught on first occurrence)	novice overload
P6	accessibility	no color-only (color+shape+text), subtitles, shake/flash/cut-in toggles, remap	colorblind/sensitive excluded

3. Wireflow

```
[Title/Main 01] --any key/single→ T [Character 02] --confirm→ [Stage/Difficulty 03] --start→ (in-run)
                                     | [Options 04] · [Meta 05] · [Leaderboard 06] · Quit
(in-run, HUD shown)
[RunStart] --1s→ [NormalRun I1]
```

```
[NormalRun] --level-up→ [Level-up I3] --confirm→ [NormalRun] (paused)
[NormalRun] --ESC→ [Pause I5] --resume→ [NormalRun]
[NormalRun] --purify threshold→ [BossWarning] --appear→ [BossFight I2]
[BossFight] --defeat→ [BossReward I4: tree→4-pick] → [NormalRun]
[4 seals broken] → [HiddenBossAttempt I2] (boss HP + time left)
(ending → result)
[HP0]→FailedWake 丿 [7:00]→DawnWake ┤→ [Result I6] [kill hidden boss before 7:00]→DreamBreak 丿
```

Wireflow — Out-game + In-game · State ↔ HUD



transition	trigger (event/condition)	to	state ↔ HUD
Title → Main	any key	Main	HUD hidden

transition	trigger (event/condition)	to	state ↔ HUD
Main → Character → Stage → run	single → confirm → start	in-run	HUD shown
NormalRun → Level-up	<code>RunTelemetry.level_reached</code> ++	I3 modal	combat paused
NormalRun → Pause	ESC/Start	I5	combat paused
NormalRun → BossWarning	<code>BossThresholdReached</code> (§12-1)	warning 5s	HUD kept
BossWarning → BossFight	<code>BossSpawned</code> (§12-1)	I2	top-center boss HP takeover
BossFight → BossReward	<code>BossDefeated</code> (§12-1)	I4	reward flow, 8s protect
4 seals → hidden boss	<code>HiddenBossSpawned</code> (§12-1)	I2 hidden	boss HP + time left
in-run → result	<code>AlarmReached</code> /HP0/ <code>DreamBreakAchieved</code> (§12-1)	I6	HUD hidden

3-1. Reference overview

Real game UI the wireframes draw from (web-harvested). **Actual images + notes are embedded inline in each screen's #### References** . Direct survivor-likes (Vampire Survivors/Brotato/20MTD) aren't on interfaceingame, so patterns come from roguelites (Hades-Slay the Spire-Returnal-RoR2), action (Hollow Knight-Destiny 2), anime (Honkai: Star Rail), with direct-genre patterns in words (§B-2).

4. Global component inventory

Visual tokens (color hex / type / spacing / motion / USS mapping) live in `lucid_dawn_ui_ux_tokens.md` . The "tokens" column below points to that file's concrete values (`ld.color.*`). Number-lock status: `lucid_dawn_ui_ux_decisions.md` .

code	component	variants	tokens (color/shape/rule)	used in
C-BAR	horizontal meter	hp-shield-xp-purge-dream-boss	horizontal only; per-type color+icon+position; danger = color+blink+border	I1-I2
C-CARD	choice card	common-uncommon-rare-lucid	rarity = color+border+corner icon; new/upgrade text badge; 1s read	I3-I4
C-RADIAL	cooldown radial	skill Q-E-ult R	radial fill + keycap glyph; ready = 1 flash + sound	I1
C-NODE	skill node	locked-available-owned	lock=lock+gray, available=focus border, owned=filled; never color-only	I4-O5
C-LIST	list/grid	menu-roster-stage-option-record	item = icon+label, selected = border+check, locked = lock+condition	O1-O2-O3-O4-O5-O6-I5
C-PANEL	info panel	char-stage-node-item-option-record	title+summary+detail; empty/error microcopy	O2-O3-O4-O5-I4
C-PROMPT	input prompt	mouse-key-pad	auto glyph for current device (no memorization)	all bottom

code	component	variants	tokens (color/shape/rule)	used in
C-BADGE	status badge	rarity-ending-combo-new	color + text label required	I3-I4-I6

5. Screen specs

Order per screen: Purpose → References → Wireframe → Legend → State matrix → Input parity → Data binding → Navigation → Edge cases → Accessibility → UX rationale → Open questions → Acceptance. Out-game O1–O6, in-game I1–I6.

5.01 TITLE — Title / Main Menu · state: menu (HUD hidden)

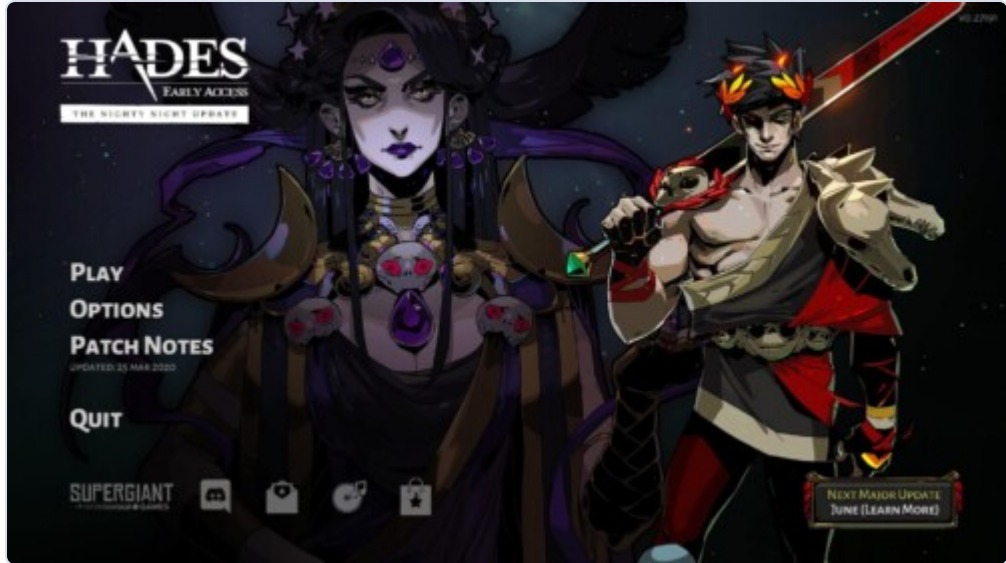
Purpose

First impression and hub to play / options / meta / leaderboard. Convey the "cute nightmare" tone and let a first-timer drop straight into single-player.

field	value
Enter	launch / Result "to Main"
Exit	single→O2 / co-op (out of scope) / O4-O5-O6 / quit
Input context	non-combat, focus
Priority	core

References — real game UI this screen draws from

Source: *ui-ref harvest* — *interfaceingame* (main menu).



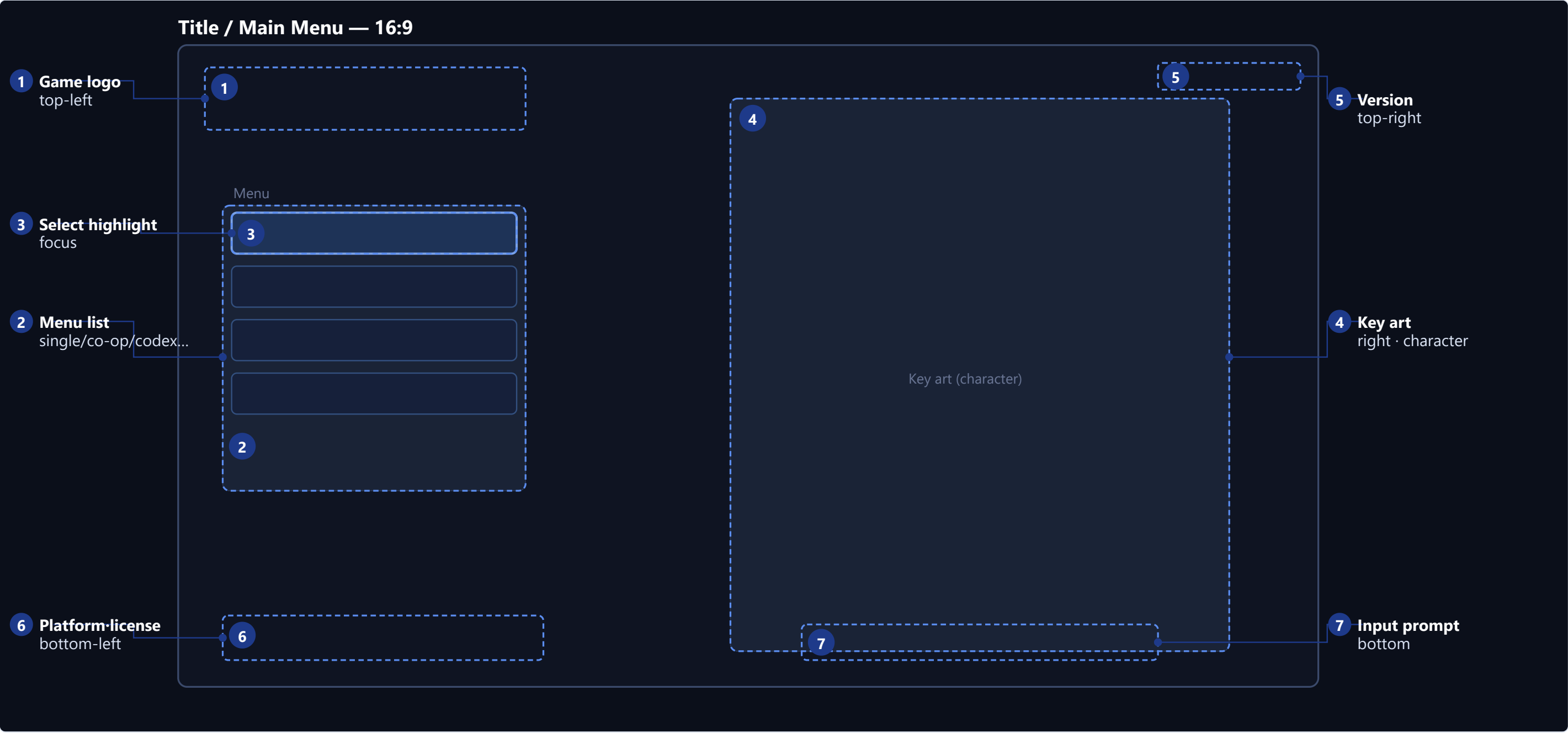
What. Logo top-left, vertical menu (Play/Options/...) center-left, character key art right, version top-right, platform/social bottom-left. **Why.** The eye flows "logo→menu" in one column and the art sells the tone. → applied: logo(1)·menu(2)·key art(4)·version(5)·platform/license(6).



What. Few menu items over a large backdrop. **Why.** Minimal items lower first-entry load. → applied: keep menu(2) short.

Words: GDD §13-3 #1 (license/official-confusion) → reserve an explicit **platform/license line (6)** on the main screen.

Wireframe



Legend

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	T1	Game logo	top-left	always	static / small motion	static asset	shown on enter	identity/tone	logo+text name
2	T2	Menu list	center-left	always	single/co-op/codex-unlock/leaderboard/options/quit; co-op locked (VS)	menu def (static)	select reacts ≤100ms	core path in one column	label text + focus
3	T3	Select highlight	menu item	always	focus emphasis + SFX	input focus	highlight on move	where am I	border+color+sound
4	T4	Key art	right	always	character illustration	static asset	—	tone/character	decorative (alt text)
5	T5	Version	top-right	always	build/version	build meta	—	build id	text
6	T6	Platform·license	bottom-left	always	platform/social/license line	static	—	prevent license confusion (§13-3 #1)	text+icon
7	T7	Input prompt	bottom	always	current-device prompt	input map (§2-5)	swaps on device change	how to operate	device glyph+text

State matrix (menu item T2/T3)

element	default	hover/focus	pressed	disabled/locked	error
Menu item (T2)	normal	emphasis+SFX, ring ≥3:1	pressed	co-op = "Coming soon" (VS lock)	"menu load error — retry"
Highlight (T3)	first item default focus	tracks	—	skips locked items	—

Input parity

action	mouse	key	pad	screen reader
move	hover	↑ / ↓	D-Pad/stick	"Single Play, 1/6"
run	click	Enter	A/○	"Single Play activate"
quit	click	—	—	"Quit, confirm required"

Data binding

code	field (GDD §)	event (GDD §12-1)	UI-proposed (needs add)	format	fallback
T2	menu/modes (static), co-op lock (§3 VS)	—	MenuFocusChanged	items	default menu
T5	build meta (static)	—	—	vX.Y	empty

Navigation

single→O2 / options→O4 / codex·unlock→O5 / leaderboard→O6 / co-op→(VS lock) / quit→confirm modal. Default focus **Single Play**.

Edge cases

Quit = irreversible → confirm modal. No save (first launch): meta/codex empty but enterable. Co-op locked: "Coming soon", not startable.

Accessibility

Focus = color+border+sound; locked = lock+text. Subtitles/text scale. License line readable contrast (§13-3 #1).

UX rationale

- **See at a glance:** logo→vertical menu in one line; only 6 items so a first-timer finds "Single Play" instantly.
- **Operate / avoid mistakes:** default focus on single; quit confirms; co-op grayed "Coming soon" prevents dead clicks.
- **Feel:** key art on the right carries the "cute nightmare" tone from the first screen.
- **First vs repeat:** novice goes single; veteran jumps to meta/leaderboard.
- **Accessibility:** lock/select not by color alone; license line legible.
- **Watch (top 3):** ① official/license confusion (§13-3 #1) → reserved license line (legal review separate); ② menu overload → few items + default focus; ③ co-op dead click → lock label.

Open questions

Hide co-op in VS vs gray it (currently gray). Title motion/sound toggle location.

Acceptance

- ☐ First launch focuses Single Play by default.
- ☐ Co-op is shown locked and never starts.
- ☐ Pad-only can move & run the menu.
- ☐ License line is present.

5.O2 CHARSEL — Character Select · state: menu

Purpose

Show how each character "wakes from the dream" differs (§5) and let the player choose. VS: Kohaku/Toko active, 4 locked. Put the evade feel up front.

field	value
Enter	Main→single / Result→Retry
Exit	confirm→O3 / back→Main
Input context	non-combat, focus

field	value
Priority	core

References

Source: ui-ref harvest — interfaceingame (character).



What. Portrait grid/list with rarity & lock state, a focused 3D character, a right info card (name/desc/stats). **Why.** Grid+lock/rarity shows ownership & playstyle at a glance; left/center/right makes "browse + compare" natural. → applied: list(2)·lock(3)·3D preview(4)·name/archetype(5)·core stats(6)·weapon/passive/ult(7).



What. Few characters as large cards with a one-line playstyle. **Why.** With a small roster, a big preview + summary aids choice. → applied: VS 2-character

big preview.

Words: the key difference here is **evade feel**, so dash distance/cooldown/safe-time/lucid-window go to the front of the stats (\$4-3).

Wireframe

Character Select (Kohaku / Toko) — 16:9

1 **Screen title**
Character Select

2 **Character list**
left · 6

3 **Select/lock mark**
color+lock

4 **3D preview**
center · rotate

1

Character list

2

3

4

3D preview

5

Info card

6

7

8

5 **Name·archetype**
recommended for

6 **Core stats**
dash·lucid window

7 **Weapon·passive·ult**
summary

8 **Start button**
bottom · Enter/A

Legend

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	E1	Screen title	top-left	always	"Character Select"	static	shown on enter	where am I	text
2	E2	Character list	left	always	6; Kohaku/Toko active, 4 locked (§3,§5)	<code>Character.id</code> (§12-2), roster(§5)	focus syncs right ≤instant	choices in one column	icon+name
3	E3	Select/lock mark	list item	always	select=border, lock=lock+gray+condition	unlock (§2-3 meta)	locked not startable	can I pick it	color+border+lock
4	E4	3D preview	center	always	model rotate/idle	<code>Character.id</code> (§12-2)	swaps ≤200ms on focus	see the look	alt text (name-playstyle)
5	E5	Name-archetype	top-right	always	name+archetype+recommended (§5)	<code>Character</code> (§5)	syncs on focus	identity one line	text
6	E6	Core stats	mid-right	always	HP-move-dash(dist/cd/safe)-lucid window	<code>Character.hp · move_speed · dash_* · lucid_window</code> (§12-2)	exact per-char values (§4-3,§5)	the evade difference	bars/radar + numbers
7	E7	Weapon-passive-ult	low-right	always	base weapon·passive-ult mod	<code>CombatWeapon · passive · ultimate_modifier</code> (§12-2)	exact per-char (§4-4,§4-6,§5)	how it fights	icon+text
8	E8	Start button	bottom	active pick	confirm → O3	routing	disabled if locked char	next step	button+prompt

Kohaku vs Toko (E6-E7 data, GDD §4-3/§5)

item	Kohaku (standard)	Toko (agile)
HP / move	110 / 6.0	90 / 6.8
dash dist/cd/safe	4.8m / 1.25s / 0.28s	4.1m / 0.85s / 0.20s
lucid window	0.10s	0.08s
base weapon	Star Pulse (single)	Phantom Needle (3-burst)
passive	Steady Waking (+10% evade reward)	Fleeting Mischief (+20% combo purify)
ult mod	Cannon +10% width	dash cd -35% after cannon
recommended	first players, stable	skilled / speedrun

State matrix (E2/E3, E8)

| element | default | hover/focus | selected | disabled/locked | loading | error | | --- | --- | --- | --- | --- | --- | | List item (E2/E3) | normal | emphasis+right sync | border+check | locked=lock+gray+"unlock cond" (§2-3) | enter fade | "char load error" | | Start (E8) | active | emphasis+ring | — | disabled+"locked character" if locked pick | — | "transition fail — retry" | | 3D preview (E4) | idle | — | — | — | "loading..." (≠ no-data) | model fail = portrait fallback |

Input parity

action	mouse	key	pad	screen reader
focus char	hover	↑ / ↓	D-Pad/stick	"Kohaku, standard, unlocked"
confirm	click	Enter	A/○	"Kohaku selected, start"

action	mouse	key	pad	screen reader
back	click	Esc	B	"Main menu"

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
E2/E3	<code>Character.id</code> (§12-2), <code>roster</code> (§5), <code>unlock</code> (§2-3)	—	<code>CharacterFocused</code> , <code>RosterLoaded</code>	list	default 2
E4	<code>Character.id</code> (§12-2)	—	<code>PreviewLoaded</code>	3D	portrait
E6	<code>Character.hp</code> · <code>move_speed</code> · <code>dash_*</code> · <code>lucid_window</code> (§12-2)	—	—	numbers/radar	"__"
E7	<code>CombatWeapon</code> · <code>passive</code> · <code>ultimate_modifier</code> (§12-2)	—	—	summary	"__"

Navigation

confirm(E8) → O3 → run(I1). back → Main. Result(I6) "Retry" enters here.

Edge cases

Locked char: focusable but E8 disabled + unlock condition. Default focus Kohaku (first-player rec, §5). Preview fail → portrait. Co-op (out of scope): per-player select / duplicates allowed (§8) unbuilt in VS.

Accessibility

Lock/select via color+border+lock; stats show numbers alongside any radar; focus announces "name:playstyle-unlock"; 3D alt text.

UX rationale

- **See at a glance:** pick on the left → center look + right performance update instantly. The core difference is evade feel, so dash/lucid-window lead the stats → Kohaku (stable) vs Toko (fast/risky) reads immediately.
- **Operate / avoid mistakes:** locked = lock+condition+disabled start; default focus on the recommended starter.
- **Feel:** 3D preview + one-line weapon/passive/ult lets you imagine the playstyle.
- **First vs repeat:** rec guides novices to stable; veterans pick Toko for speedrun builds.
- **Accessibility:** not color-only; numbers beside radar.
- **Watch:** ① choosing blind → evade stats up front + one-line summary; ② lock confusion (colorblind) → lock+text; ③ radar imprecise → numbers; ④ license/official confusion (§13-3 #1) → name-display guideline (legal separate).

Open questions

Unlock-condition text exposure. Stats radar vs bars (bars are precise for the 4 evade stats).

Acceptance

- ☐ Kohaku/Toko active, 4 locked, distinguishable.

- ☐ Preview & stats sync $\leq 200\text{ms}$ on focus.
- ☐ Dash dist/cd/safe/lucid-window shown per GDD values.
- ☐ Colorblind mode distinguishes lock/select.
- ☐ Locked char disables Start.

5.O3 STAGESEL — Stage / Difficulty Select · state: menu

Purpose

Pick the stage (VS: LD-001 Sleeping Room) and difficulty (Sleepy–Lucid), and show what each choice changes (foes/boss/reward/purify requirement).

field	value
Enter	after O2 confirm
Exit	start→run(l1) / back→O2
Input context	non-combat, focus
Priority	core

References

Source: ui-ref harvest — *interfaceingame (map/level)*.



What. Progress nodes/paths + a reward preview per node. **Why.** Show what's coming (foes/reward) before choosing. → applied: stage list(2)·info(4).



What. Current vs locked regions on a map. **Why.** Convey unlock state spatially. → applied: lock/select mark(3).

Words: difficulty(5) changes not just HP but purify requirement · threat · reward multipliers (\$2-4) → info panel updates with difficulty.

Wireframe

Stage / Difficulty Select — 16:9

1 Screen title
Stage Select

2 Stage list
LD-001 + locked

3 Lock/select mark
unlock cond.

Stage list

Stage info

Difficulty

4 Stage info
theme·foes·boss·reward

5 Difficulty
Sleepy~Lucid

6 Mode summary
20min · goal

7 Start button
bottom · input

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	G1	Screen title	top-left	always	"Stage Select"	static	instant	where	text
2	G2	Stage list	left	always	LD-001 active + 4 locked (§6-1)	stage def(§6-1), unlock	focus syncs right	where I fight	thumb+name
3	G3	Lock/select mark	item	always	select=border, lock=lock+condition	unlock cond(§6-1)	locked not startable	can I go	color+lock+text
4	G4	Stage info	top-right	always	theme-foes-boss-reward (§6-2)	stage(§6-2), diff mult(§2-4)	updates on focus/difficulty	what shows up	text+icon
5	G5	Difficulty	mid-right	always	Sleepy/Normal/Nightmare/Lucid + multipliers	diff mult(§2-4)	changing G5 updates G4	choose challenge	segmented+numbers+color
6	G6	Mode summary	right	always	Core Dream Run-20min-goal(survive/Dream Break)	mode(§2-4)	—	what I do	text
7	G7	Start button	bottom	active pick	confirm → run	routing	disabled if locked	start	button+prompt

Difficulty data (§2-4)

difficulty	purify req	enemy HP	enemy threat	reward
Sleepy	0.85x	0.85x	0.80x	0.90x
Normal	1.00x	1.00x	1.00x	1.00x
Nightmare	1.18x	1.20x	1.25x	1.15x
Lucid	1.35x	1.35x	1.45x	1.30x

State matrix (G3, G5, G7)

element	default	hover/focus	selected	disabled/locked	error
Stage (G3)	normal	emphasis+sync	border	locked=lock+"unlock cond"(§6-1)	"stage load error"
Difficulty (G5)	Normal default	emphasis	selected+multipliers	Lucid etc. may be meta-locked	—
Start (G7)	active	emphasis+ring	—	disabled if locked stage	"transition fail"

Input parity

action	mouse	key	pad	screen reader
focus stage	hover	arrows	D-Pad	"Sleeping Room, unlocked"
change difficulty	click	←/→	LB/RB	"Normal, reward 1.0x"
start	click	Enter	A/○	"Sleeping Room, Normal start"

action	mouse	key	pad	screen reader
back	click	Esc	B	"Character select"

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
G2/G3	stage(§6-1), unlock	—	StageFocused , StagesLoaded	list	LD-001
G4	stage(§6-2), diff mult(§2-4)	—	StageInfoUpdated	text/icon	default
G5	diff mult(§2-4)	—	DifficultyChanged	multipliers	Normal

Navigation

start(G7) → run(l1, RunStart). back → O2. Non-LD-001 stages active only when §6-1 unlock met.

Edge cases

VS plays LD-001 only; others locked preview. Lucid difficulty may be meta-locked (lock+condition). Changing difficulty must update info numbers immediately (avoid false expectation).

Accessibility

Difficulty difference via multiplier numbers + bars, not color only; locked = lock+condition; info text scalable.

UX rationale

- **See at a glance:** pick stage left → theme/foes/boss/reward right; changing difficulty updates those numbers so "what gets harder" is visible.
- **Operate / avoid mistakes:** locked stage/difficulty = lock+condition + disabled start.
- **Feel:** locked stages still preview as a "next goal".
- **First vs repeat:** Normal-LD-001 default for novices; veterans push Nightmare/Lucid for reward.
- **Accessibility:** difficulty conveyed as numbers too.
- **Watch:** ① not knowing what difficulty changes → multipliers in info (§2-4); ② lock confusion → lock+condition; ③ late-game tedium (§13-3 #3) → mode summary states the Dream Break goal for motivation.

Open questions

Difficulty lock condition (meta). Stage list map vs cards (VS = 1 stage → cards fine).

Acceptance

- ☐ LD-001 active, others locked preview.
- ☐ Difficulty change updates info multipliers immediately.
- ☐ Colorblind mode distinguishes difficulty/lock.

- ☐ Pad-only stage/difficulty select & start.

5.O4 SETTINGS — Options / Settings · state: menu (or from Pause)

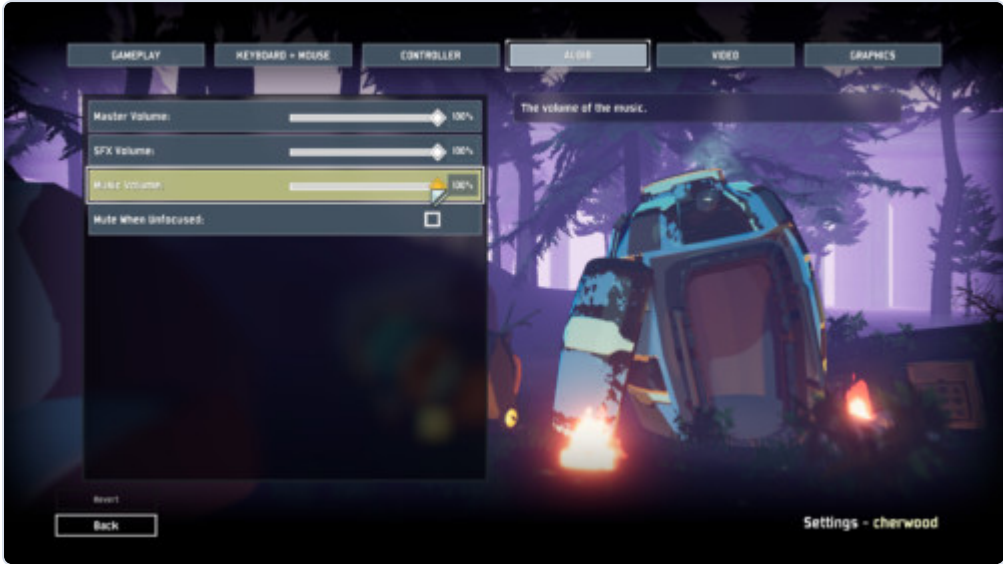
Purpose

Group gameplay/display/audio/UI-accessibility/controls options (incl. GDD §9-3 accessibility) and explain what each changes. **Accessibility is a first-class citizen.**

field	value
Enter	Main(O1) / Pause(I5)
Exit	back → caller
Input context	non-combat, focus
Priority	core (accessibility)

References

Source: *ui-ref harvest — interfaceingame (settings)*.



What. Top tabs (Gameplay/Keyboard/Controller/Audio/Video/Graphics), left sliders/toggles, right description, bottom Back/Revert. **Why.** Tabs + left control + right description is the low-learning standard. → applied: tabs(1)·options(2-3)·description(4)·bottom(6).



What. Per-item current value + left/right adjust. **Why.** Slider/stepper makes value changes clear. → applied: option control(3).

Words: highlight a dedicated **accessibility group(5)** — colorblind palette · screenshake · flash · post-processing · hit-judgment display · cut-in simplify · auto-aim bias · lucid-evade assist · subtitles (§9-3).

Wireframe

Options / Settings (tabs + options + desc) — 16:9

1 **Tab row**
top · categories

3 **Option control**
slider/toggle

2 **Option list**
sliders·toggles

5 **Accessibility group**
colorblind·shake·subs

1 Tabs: Gameplay·Display·Audio·UI/Access·Controls

Option list

2

3

5

Description panel

4

4 **Description panel**
right · explain

6

6 **Apply·revert·back**
bottom · input

Legend

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	S1	Tab row	top	always	Gameplay·Display·Audio·UI/Access·Controls	categories (static)	tab swaps options instantly	group to find	tab label + selected
2	S2	Option list	left	always	per-tab sliders/toggles/steppers	settings (saved)	change reflects/previews	all in one place	label+value text
3	S3	Option control	row	always	slider/toggle; current value	settings (saved)	adjust ≤100ms reflect	clear value change	value text+steps
4	S4	Description panel	right	focus option	effect of the focused option	option meta (static)	desc updates on focus	what it changes	text
5	S5	Accessibility group	left (section)	UI/Access tab	colorblind-shake-flash-cut-in-subst-aim-assist (§9-3)	a11y settings (saved)	toggle applies at once	a11y front and center	toggle+state text
6	S6	Apply-revert-back	bottom	always	apply/cancel/exit	—	apply confirms save	safe changes	button+prompt

State matrix (S3)

element	default	hover/focus	active	disabled	error
Option control (S3)	current value	emphasis+description	dragging/toggling	dependent disabled+reason ("fullscreen only")	"settings save failed — retry"
Tab (S1)	first tab	emphasis	pressed	—	—

Input parity

action	mouse	key	pad	screen reader
switch tab	click	Q/E·Tab	LB/RB	"Audio tab"
focus option	hover	↑ / ↓	D-Pad	"Music volume, 80%"
adjust value	drag/click	←-/→	stick/D-Pad	"Music volume 75%"
apply/back	click	Enter/Esc	A/B	"Applied"

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
S2/S3	settings (saved; §9-3 a11y items)	—	<code>SettingChanged</code> , <code>SettingsLoaded</code>	value	default
S5	a11y settings (§9-3)	—	<code>AccessibilityChanged</code>	toggle/value	default off/standard
S6	—	—	<code>SettingsApplied</code> , <code>SettingsReverted</code>	—	—

Navigation

back → caller (Main or Pause). From Pause, combat stays paused.

Edge cases

Dependent options (fullscreen↔resolution) disabled+reason. Unapplied change on back: "unsaved changes — apply/cancel". Remap conflict: same-key warning + resolution.

Accessibility

This screen is the gateway to all of it: colorblind palette, shake/flash/post-processing/cut-in toggles, hit-judgment display, auto-aim bias, lucid-evade assist, subtitles (size/bg/speed) — all §9-3. Toggle state = color + text (ON/OFF).

UX rationale

- **See at a glance:** tabs group options; the focused option's description on the right answers "what does this do?".
- **Operate / avoid mistakes:** apply/revert make changes safe; dependent options disable with a reason; remap conflicts warn.
- **Feel:** clarity over flourish; volume/shake preview immediately.
- **First vs repeat:** accessibility/controls up front for those who need them; veterans go to graphics/gameplay.
- **Accessibility:** this screen exposes all of §9-3; toggle = color+text.
- **Watch:** ① excluding colorblind/sensitive users (§13-3 #5) → colorblind/shake/flash/cut-in toggles first-class; ② lost changes → unsaved-change confirm; ③ key conflict → warn+resolve.

Open questions

Accessibility presets (colorblind/low-stim bundle). Which options apply live in-game (shake etc.).

Acceptance

- ☐ All §9-3 accessibility items exposed; toggles shown by color+text.
- ☐ Focusing an option updates the description.
- ☐ Back with unsaved changes shows a confirm modal.
- ☐ Pad-only tab/value/apply works.

5.O5 META — Meta Progress (permanent unlocks / codex) · state: menu

Purpose

Between runs, spend currency (dream shards/stardust/lucid core, §2-3) to **unlock by play only** (characters/skill trees/ult/relics/difficulty) and view a codex. **GDD §10 BM: no paid power, no gacha → no purchase UI.**

field	value
Enter	Main(O1) "Codex/Unlock"
Exit	back → Main
Input context	non-combat, focus

field	value
Priority	core (repeat motivation)

References

Source: ui-ref harvest — interfaceingame (skill tree / collection).



What. Upgrade rows (name+effect+level pips+MAX), currency top-right, selected item desc bottom-right. **Why.** Rows+currency+desc make "what costs

what" clear. → applied: unlock list(3)·item(4)·currency(2)·desc(5)·unlock button(6).



What. Collected items/cards grid. **Why.** Codex collection progress. → applied: codex tab (C-LIST).

Words: **no cash purchase button/price** — every unlock spends play-earned currency only (\$10). This is the deliberate difference from a typical shop screen.

Wireframe



Legend

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	K1	Title-tabs	top-left	always	Permanent upgrades / Codex / Records	categories (static)	tab swaps instantly	grouping	tab label
2	K2	Currency	top-right	always	dream shards-stardust-lucid core (§2-3)	meta currency (§2-3)	reflects on change	what I can spend	icon+number
3	K3	Unlock list	center	upgrade tab	rows: name-effect-state(owned/locked)	unlock items(§2-3), <code>Item</code> / <code>SkillNode</code> (§12-2)	focus syncs desc	what can I open	row label+state
4	K4	Item state	row	always	locked / available / owned	cost:prereq(§2-3)	"available" when currency met	can I open now	lock/check+color
5	K5	Item desc-cond	right	on focus	effect-cost-unlock condition	item meta	updates on focus	confirm before unlock	text
6	K6	Unlock button	low-right	available	spend currency to unlock (no cash)	cost(§2-3)	click deducts+confirm	confirm unlock	button+prompt
7	K7	Back-input	bottom-left	always	exit	—	—	leave	button

State matrix (K4, K6)

element	default	hover/focus	selected	disabled/locked	error
Item (K4)	owned=check / not=outline	emphasis+desc	just-unlocked mark	prereq unmet=lock+condition / low currency="N short"	"load error — retry"
Unlock (K6)	active when available	emphasis+ring	—	disabled+reason if short/unmet	"unlock failed — retry"

Input parity

action	mouse	key	pad	screen reader
switch tab	click	Q/E	LB/RB	"Codex tab"
focus item	hover	arrows	D-Pad/stick	"Lucid Sense, cost 50 shards, lock: clear boss once"
unlock	click	Enter	A/○	"unlocked, 50 shards spent"
back	click	Esc	B	"Main"

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
K2	meta currency(§2-3)	—	<code>CurrencyChanged</code>	number	0
K3/K4	unlock items(§2-3), <code>SkillNode</code> / <code>Item</code> (§12-2)	—	<code>UnlockablesLoaded</code> , <code>UnlockStateChanged</code>	list	locked
K5	item meta-cost(§2-3)	—	<code>UnlockableFocused</code>	text	"select"

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
K6	cost(\$2-3)	—	Un`lockPurchased (currency)	—	—

No cash-payment event is defined (§10 BM).

Navigation

back → Main(O1). Unlocked characters/difficulty become active in O2/O3 at once. Codex tab shows enemy/boss/item collection (§6-3, §4-5, §10).

Edge cases

Currency short: button disabled+amount. Prereq unmet: lock+condition ("clear boss once"). Unlock confirm (currency spend is irreversible). Empty codex entry: "undiscovered" (minimize spoilers).

Accessibility

Unlock state (locked/available/owned) via color+lock/check; currency/cost numbers explicit; description scalable; no time pressure.

UX rationale

- **See at a glance:** center unlock list, right shows the focused item's effect/cost/condition; currency top-right tells "what I can open for how much".
- **Operate / avoid mistakes:** short currency or unmet prereq disables with a reason; unlock confirms.
- **Feel:** unlocking fills items and the codex grows — long-term repeat motivation.
- **First vs repeat:** highlight 1–2 next unlocks for novices; veterans optimize the tree.
- **Accessibility:** state not color-only.
- **Watch:** ① **monetization confusion (§10 BM)** → never any cash UI, currency unlocks only; ② late-game tedium (§13-3 #3) → codex/unlock as long-term goals; ③ colorblind lock confusion → lock+condition.

Open questions

Codex spoiler policy (silhouette vs hidden). Visual distinction between permanent tree and in-run tree (I4).

Acceptance

- ☐ No cash-purchase UI; all unlocks spend currency.
- ☐ Short/unmet items show disabled+reason.
- ☐ Unlock deducts currency and reflects in O2/O3.
- ☐ Colorblind mode distinguishes locked/owned.

Purpose

Show GDD \$10 records (Dream Break Time, boss splits, hidden-boss unlock time, no-damage, highest purify, co-op, character clears) split by stage/difficulty/character/solo-co-op/condition, and highlight **my rank** to drive competition.

field	value
Enter	Main(O1) "Leaderboard"
Exit	back → Main
Input context	non-combat, focus, network-dependent
Priority	extended (competition)

References

Source: no direct leaderboard capture on the harvest games; rank-table composition derived from a result stats panel.



What. A left stats panel listing per-item numbers line-aligned. **Why.** Line-aligned tables compare many numbers fastest. → applied: rank table(4)-my-rank highlight(5) row structure.

Words: a real leaderboard capture can be added later from Game UI Database **Leaderboards & Ranking** (scrn=55) — see Appendix B (note the GUIDB headless limit). Filter(2)-record type(3)-my-rank(5) are derived from common leaderboard patterns in words.

Wireframe

Leaderboard / Records — 16:9

1 Screen title
Leaderboard

2 Filter tabs
difficulty·char...

3 Record type
DB time·purify...

1

2

3

Record type

4

Rank table

5

4

Rank table
rank·record·char

5

My rank highlight
highlight

6

Back · input
bottom

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	L1	Screen title	top-left	always	"Leaderboard"	static	instant	where	text
2	L2	Filter tabs	top	always	stage-difficulty-character-solo/co-op-condition (§10)	filters (§10 axes)	change refreshes table	compare like-for-like	tab label+selected
3	L3	Record type	left	always	DB time-boss split-hidden unlock-no-damage-highest purify (§10)	record types (§10)	select refreshes table	what we compete	label+selected
4	L4	Rank table	center	always	rank-player-record-character	record data (§10)	sorted after load	rank compare	table header+values
5	L5	My-rank highlight	table row	my record exists	highlight + jump-to	my record (§10)	scrolls to my row on enter	my position	color+border+"YOU"
6	L6	Back-input	bottom	always	exit	—	—	leave	button

State matrix (L4, L5)

element	default	hover/focus	disabled	loading	empty	error
Table (L4)	sorted	row emphasis	offline = local-only + "Offline"	"loading ranks..." (network)	"no records — be the first"	"load failed — retry"
My rank (L5)	highlight	—	no record = hidden + "no record"	—	—	—

Input parity

action	mouse	key	pad	screen reader
filter/type	click	Q/E-arrows	LB/RB-D-Pad	"Difficulty: Normal, record: Dream Break Time"
scroll	wheel/drag	↑ / ↓	stick	"1st 9:12 Toko"
to my rank	button	Home	Y	"jump to my rank 14th"
back	click	Esc	B	"Main"

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
L2/L3	axes-record types(§10)	—	LeaderboardFilterChanged	tabs	default
L4	records(§10), dream_break_stage_times (§12-2)	—	LeaderboardLoaded	table	empty/error
L5	my record(§10)	DreamBreakAchieved (§12-1)	MyRankResolved	row	hidden

Navigation

back → Main(O1). Result(I6) "Leaderboard" enters here with the matching record filter.

Edge cases

Offline: local-only + "Offline" badge, resync on reconnect. No records: "be the first". Loading: skeleton + "loading..." (≠ enter anim). Cheat-report/filters later.

Accessibility

My rank via color + "YOU" label + border; sticky table header; network state in words; scalable text.

UX rationale

- **See at a glance:** filter first (so comparisons are like-for-like), then record type, then table; on enter it scrolls to my rank.
- **Operate / avoid mistakes:** offline/loading/empty stated in words so "why is it blank?" never happens.
- **Feel:** my-rank highlight + new-record markers drive the next attempt.
- **First vs repeat:** "be the first" lowers novice pressure; veterans chase Dream Break Time/no-damage.
- **Accessibility:** my row marked beyond color.
- **Watch:** ① blank-on-network-fail confusion → offline/error/loading distinct copy; ② unfair mixed comparison → force filter first; ③ colorblind my-rank id → "YOU" label.

Open questions

Co-op record exposure (VS = solo). Cheat-report/verification policy. Season/weekly reset.

Acceptance

- ☐ Filter/record change refreshes the table immediately.
- ☐ On enter, scrolls to and highlights my rank.
- ☐ Offline/loading/empty shown with distinct copy.
- ☐ Colorblind mode identifies my rank by label.

5.11 HUD-IN — In-game HUD (combat) · state: NormalRun

Purpose

Let the player read **time left** · **survival** · **resources** · **progress** without moving their eyes, and judge move/dodge/skill/ult. Priority #1: the HUD never hides bullets/telegraphs even in chaos.

field	value
Enter	<code>RunStart</code> +1s → <code>NormalRun</code> (\$4-1)
Exit	level-up(I3)/reward(I4)/pause(I5) overlay / ending → I6
Input context	real-time combat

field	value
Priority	all core (lucid combo extended) — §9-2

References

Source: ui-ref harvest — interfaceingame (HUD).



clear center, minimal chrome; vitals top-left (2·3·4·5), skills/ult bottom (8·9).

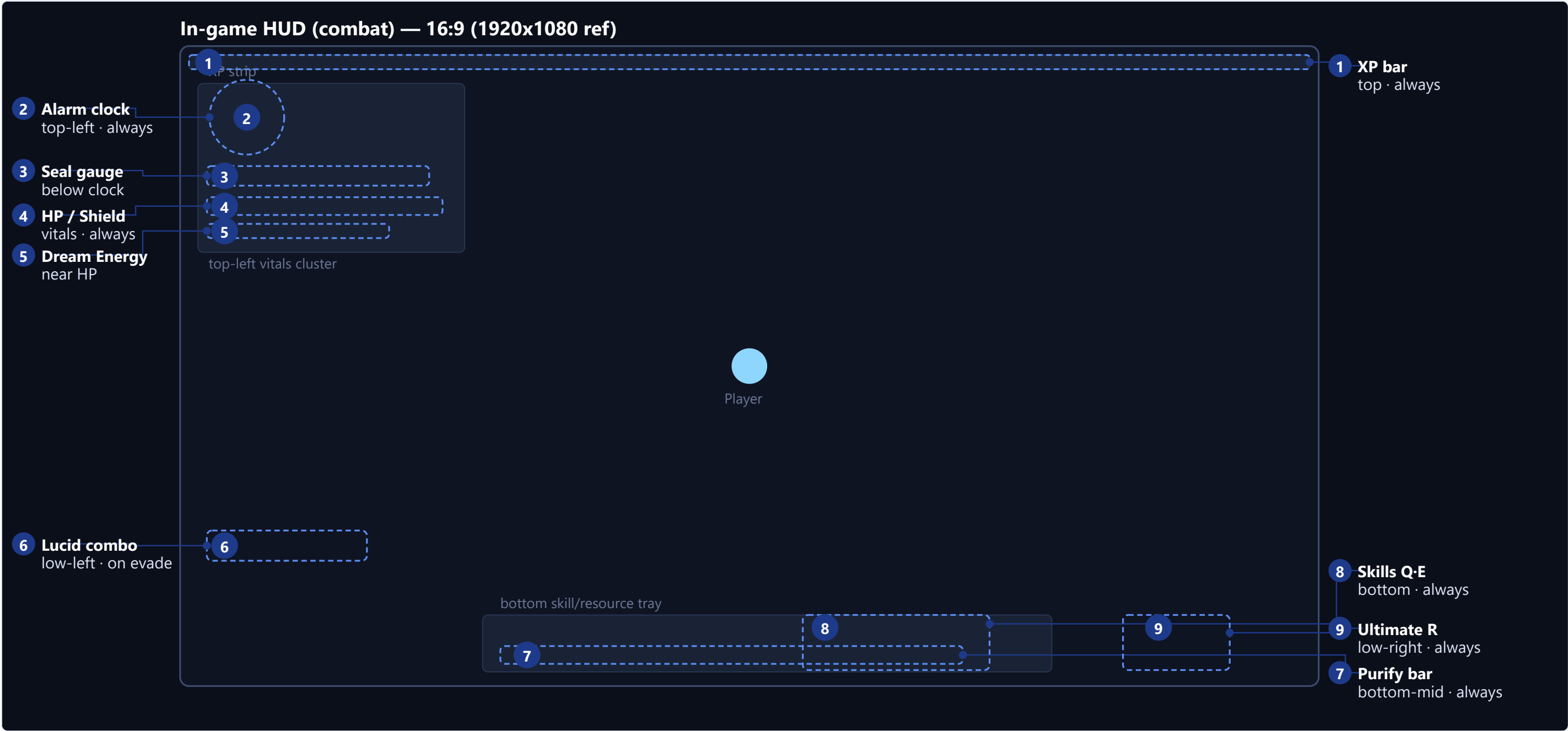
What. Minimal chrome, top-left HP/resource, bottom-right ability. **Why.** Push HUD to the edges and keep center clear for combat legibility. → applied:

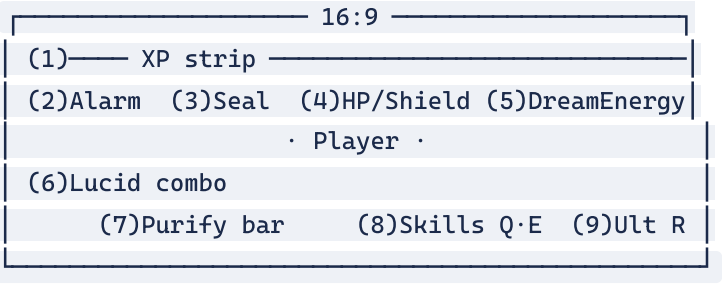


What. Resource/cooldown shown as radials/gauges. **Why.** Read resource/cooldown without text. → applied: skill radial(8)·ult gauge(9).

Words: timers usually sit **top-center/top-right**, but here it's an **alarm clock (diegetic) fixed top-left** (deliberate exception — see UX rationale). XP = a thin full-width strip at the very top.

Wireframe





Legend

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	A1	XP bar	top full-width	always	fill 0→100%; flash+reset on level-up	XP (§4-5), <code>RunTelemetry.level_reached</code> (§12-2); event UI-proposed	reflects ≤200ms	progress to next growth	fill+number, not color-only
2	A2	Alarm clock	top-left	always	hand sweep; 1140–1200s edge brighten + hand emphasis (§4-1)	<code>run.timer</code> 0–1200s(§4-1), <code>AlarmReached</code> (§12-1)	time-left read ≤1s anytime (§3-2)	top rule visibility	clock+digits, final-60s sound+brightness
3	A3	Seal gauge	below clock	always	4 seals; lights on boss kill	seal slot(§4-2), <code>BossDefeated</code> (§12-1)	lights ≤300ms on break	progress to early clear	pips+lock/unlock icon
4	A4	HP/Shield	top-left	always	drop+red blink on hit; shield distinct color; low-HP border	<code>Character.hp</code> (§12-2), shield (§12-2 add)	hit reflects ≤100ms; low-HP (≤25% [TBD]) color+blink+border	near-death by shape too	color+blink+border+number
5	A5	Dream Energy	near HP	always	charges via evade/kill; full = ult ready signal	dream energy(§4-3-§4-6; field §12-2 add), <code>EvasionLucid</code> / <code>EvasionNearMiss</code> (§4-3)	full → signal with A9 at once	foreshadow ult timing	fill+number+icon
6	A6	Lucid combo	low-left	6s after evade	combo 1→3 + multiplier; fades if no streak in 6s	<code>LucidComboStep</code> (§4-3), max 3(§4-3)	updates ≤0.2s; holds 6.0s (§4-3)	reward the dodge rhythm	step+color+sound
7	A7	Purify bar	bottom-center	always	fills to slot threshold; boss warning at threshold	purify(§4-2 thr 260/620/1080/1640), <code>PurgeGained</code> · <code>BossThresholdReached</code> (§12-1)	gain ≤200ms; threshold→warning at once	distance to next boss	fill+number+source toast
8	A8	Skills Q·E	bottom	always	cooldown radial; ready flash; pressed+SFX	skill cd(§2-5; field §12-2 add)	ready flash ≤200ms after cd	usable? how long?	radial+keycap+sound
9	A9	Ult gauge R	low-right	always	lit at dream energy 100; cinematic on use	dream/100(§4-6), R(§2-5)	lit at 100 at once; input ≤100ms	finisher resource apart	fill+number+lit

State matrix (key dynamic elements)

element	default	pressed/active	disabled/locked	loading	empty	error
Skill (A8)	radial fill	pressed+SFX, radial 0	gray+lock+"unlocked later" (pre-tree)	enter fade	N/A	"cd error — default"
Ult (A9)	charging fill	cinematic on fire	dim+remaining if <100	enter fade	N/A	resource missing = last value+border

element	default	pressed/active	disabled/locked	loading	empty	error
HP/Shield (A4)	current	red blink on hit	N/A	enter fill	HP0 = FailedWake transition	missing = last value+border
Lucid combo (A6)	hidden	appears+increments on evade	N/A	appear anim	fades after 6s	N/A
Alarm (A2)	normal	60s warning: brightness+hand	N/A	enter set	N/A	missing = "syncing..." + last

Input parity

action	key	pad	screen reader
move	WASD	left stick	"move"
dash/dodge	Space	A/○	"dodge, cd n s"
skill 1-2	Q-E	LB·RB	"skill1 ready/cd"
ultimate	R	Y/△	"ult ready/charge n%"
camera/ping	hotkey	right stick/D-Pad	"ping"

Data binding (event-driven)

code	field (GDD §)	event (GDD §12-1/§4-3)	UI-proposed (needs add)	format	fallback
A1	XP(§4-5); <code>level_reached</code> (§12-2)	—	<code>XpChanged</code> , <code>LevelReached</code>	0–100%+Lv	0%
A2	<code>run.timer</code> 0–1200s(§4-1)	<code>AlarmReached</code> (§12-1)	<code>RunTimerTick</code>	mm:ss left	0:00
A3	seal(§4-2); <code>Boss.seal_slot</code> (§12-2)	<code>BossDefeated</code> · <code>BossThresholdReached</code> (§12-1)	<code>SealStateChanged</code>	4 pips	0
A4	<code>Character.hp</code> (§12-2); shield (§12-2 add)	—	<code>HpChanged</code> , <code>ShieldChanged</code>	n/max	last
A5	dream energy(§4-3·§4-6; field §12-2 add)	<code>EvasionLucid</code> · <code>EvasionNearMiss</code> (§4-3)	<code>DreamEnergyChanged</code>	n/100	0
A6	combo step(§4-3; field §12-2 add)	<code>LucidComboStep</code> (§4-3)	—	x1–x3	hidden
A7	purify(§4-2; field §12-2 add)	<code>PurgeGained</code> · <code>BossThresholdReached</code> (§12-1)	<code>PurgeGaugeChanged</code>	n/thr	0
A8	skill cd(§2-5; field §12-2 add)	—	<code>SkillCooldownChanged</code> , <code>SkillReady</code>	s left	ready
A9	dream/100(§4-6)	—	<code>UltimateChargeChanged</code> , <code>UltimateReady</code>	n/100	0

UI events/fields — GDD additions needed: runtime state `RunState{timer, purge_current, seal_slot_index}` , `PlayerRuntime{hp_current, shield_current, dream_energy, ultimate_charge, lucid_combo_step}` , `SkillsSlot{cooldown_remaining}` are not in §12-2. **XP bar isn't in §9-2 either** → UI-proposed. We do not claim these are bound to §12-1 events.

Navigation

No routing (display only). State transitions only: `NormalRun` →I3/I5/I2/I6.

Edge cases

Concurrent ends (§4-1): 7:00 + boss kill → `DawnWake` priority (no boss reward); level-up open at 7:00 → close window then `DawnWake` . Ult cinematic at 7:00 → cut/fade then `DawnWake` . Data flood: toasts merge/throttle. Missing value: last value + border.

Accessibility

Colorblind palette distinguishes enemy bullets / ally bullets / hazard zones by shape+color (§9-3). Shake/post-processing strength, hit-judgment display (§9-3). Low-HP/threshold = color+blink+sound (triple).

UX rationale

- **See at a glance:** first read ① time to 7:00 (top-left alarm), ② HP (top-left), ③ purify to next boss (bottom). Action resources sit bottom/low-right where the hand is. Center is empty so bullets/telegraphs/character stay visible.
- **Operate / avoid mistakes:** skills/ult show "usable now?" as radial fill + key glyph — no memorization. Low resource = dim + remaining.
- **Feel:** lucid evade pops the combo instantly; full dream energy signals on both the resource bar and the ult slot. Juice is capped so it never buries info.
- **First vs repeat:** unfamiliar info (combo/purify) appears on first occurrence (combo hidden otherwise); veterans read fixed positions fast.
- **Accessibility:** hazard/low-HP/threshold never color-only.
- **Watch:** ① danger buried in volume (§13-3 #5) → HUD to edges, center clear, hazards by shape too; ② missing time because alarm is top-left → final-60s edge brighten+sound (boss-fight handling in I2); ③ juice hiding info → caps+toggles; ④ purify/seal/dream confusion → distinct color+position+icon, labeled meaning.

Open questions

Low-HP threshold % (currently 25%). Promote XP bar to §9-2. Verify purify(A7) vs seal(A3) meaning overlap.

Acceptance

- ☐ A random screenshot lets you read time-to-7:00 within 1s.
- ☐ At 180 enemies, player/lethal bullets/zones stay distinct in colorblind mode.
- ☐ Hit reflects HP drop within 100ms via color+blink+border.
- ☐ At dream 100 the resource bar and ult slot signal together.
- ☐ Pad-only performs all combat actions.

5.I2 HUD-BOSS — In-game HUD (boss / hidden boss) · state: `BossFight` · `HiddenBossAttempt`

Purpose

Add **boss HP** · **phase** · **shield-crack/groggy** · **counter window** without losing survival/time info. Key: the **top-center slot is taken over by the boss HP bar**.

field	value
Enter	BossSpawned / HiddenBossSpawned (§12-1)
Exit	BossDefeated →I4 / AlarmReached →DawnWake / DreamBreakAchieved →I6
Input context	real-time (boss + weakened waves 45%/hidden 25% — §6-4)
Priority	core

References

Source: ui-ref harvest — interfaceingame; boss-bar takeover from words (no clean boss-bar capture this run).



What. Enemy/threat status emphasized separately from the player HUD. **Why.** Surface "the chance to break it". → applied: shield-

crack/groggy(11)·counter window(12).

Deliberate exception (important): the survivor-like standard is "timer top-center → boss HP takes that slot". Here the timer is an **alarm clock (a world object), fixed top-left**. Instead the **top-center slot is left empty in normal play and owned by the boss HP bar in a fight** (same physical slot shared over time). Safeguards: ① the alarm clock stays top-left during boss fights; ② in a hidden-boss fight, per §4-1 the **time-left is inset into the boss bar**. Boss-bar = top-center, segmented at HP55% phase (§7-1), with the boss NAME (LoL Swarm / GUIDB pattern, §B-2).

Wireframe

In-game HUD (boss / hidden boss) — top-center takeover

10 Boss HP bar

top-center · segmented

2 Alarm clock

top-left · kept

3 Seal→Hidden

below clock

2

3

clock stays top-left in boss fight

top-center slot (empty → boss owns)

10

11

13

13 Time inset

hidden boss only

11 Crack/Groggy

under boss HP



Player

12

Seal trial banner (during trial)

14

12 Counter window

0.7s after evade

14 Trial banner

during trial

7

7 Purify bar

30% in boss

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
10	A10	Boss HP bar	top-center	boss/hidden	horizontal fill; HP55% phase segment (§7-1); boss name	<code>Boss.hp</code> · <code>phase_rules</code> (§12-2), <code>BossSpawned</code> (§12-1)	hit ≤100ms; phase flips at 55%	threat/remaining always on top	fill+segment line+number
2	A2	Alarm clock	top-left	kept in boss fight	as I1	<code>run.timer</code> (§4-1)	time read ≤1s kept	time pressure kept	as I1
3	A3	Seal→Hidden	below clock	always	at 4 seals the gauge becomes the hidden-boss gauge (§4-2)	<code>seal/hidden</code> (§4-2), <code>HiddenBossSpawned</code> (§12-1)	converts at 4 seals	stage transition clear	shape change+label
11	A11	Shield-crack/Groggy	under boss HP	boss	crack accrues; 100→4s groggy (×1.35 dmg) (§7-1)	<code>BossPattern.shield_crack_value</code> · <code>groggy_interaction</code> (§12-2), <code>EvasionBossLucid</code> (§4-3)	crack ++ on boss-lucid counter; 100→groggy 4.0s	the chance to break it	separate bar+groggy icon+sound
12	A12	Counter window	near player	0.7s after lucid evade	brief lit; invites counter	<code>LucidCounterWindowOpen</code> (§12-1), 0.7s(§4-3)	lit at once on evade, 0.7s	counter timing feel	lit+sound+color/shape
13	A13	Time inset	inside boss HP	hidden boss only	time-left inset in boss bar	<code>run.timer</code> (§4-1), <code>HiddenBossSpawned</code> (§12-1)	inset shown at once on hidden	time+HP at once	digits+bar, dual with top-left clock
14	A14	Seal-trial banner	mid-bottom	during trial(§7-2)	trial name-time limit-goal; countdown	<code>DreamBreakTrial.id</code> · <code>duration_limit</code> · <code>success_condition</code> (§12-2)	shown at trial start; 1s ticks	trial goal/time clear	text+countdown+color
7	A7	Purify bar	bottom-center	boss	only 30% of normal-enemy purify (§4-2 BossActive)	<code>purify</code> (§4-2), <code>PurgeGained</code> (§12-1)	30% ratio in boss	keep progress context	fill+number

State matrix (boss additions)

element	default	active	groggy/special	locked	loading	error
Boss HP (A10)	fill+name	hit flash	groggy: contrast+icon+×1.35 mark	hidden when not boss (empty slot)	enter slide	missing = last+border
Shield-crack (A11)	0	++ on counter	100→groggy flip	N/A	—	missing = last
Counter window (A12)	hidden	0.7s lit after evade	—	no judgment during ult i-frames (§4-3)	—	N/A
Time inset (A13)	hidden (normal boss)	shown in hidden fight	60s warning emphasis	—	enter	missing = fall back to top-left clock

Input parity

As I1 (shared combat). Additionally **counter** fires within 0.7s after a lucid evade (§4-4 lucid counter 28). Same pad/key.

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
A10	<code>Boss.hp</code> · <code>phase_rules</code> · <code>pattern_list</code> (§12-2)	<code>BossSpawned</code> (§12-1)	<code>BossHpChanged</code> , <code>BossPhaseChanged</code>	n/max+phase	last

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
A11	<code>BossPattern.shield_crack_value · groggy_interaction</code> (§12-2)	<code>EvasionBossLucid</code> (§4-3)	<code>ShieldCrackChanged</code> , <code>GroggyChanged</code>	n/100	0
A12	0.7s window (§4-3)	<code>LucidCounterWindowOpen</code> (§12-1)	—	on/off	off
A13	<code>run.timer</code> (§4-1)	<code>HiddenBossSpawned</code> (§12-1)	<code>RunTimerTick</code>	mm:ss	top-left clock
A14	<code>DreamBreakTrial.*</code> (§12-2)	—	<code>TrialStarted</code> , <code>TrialTick</code> , <code>TrialResult</code>	name+countdown	hidden

Navigation

`BossFight` →(kill)→I4 / (HP0)→FailedWake(I6) / (7:00)→DawnWake(I6). `HiddenBossAttempt` →(kill)→DreamBreak(I6) / (7:00)→DawnWake.

Edge cases

Dream Break cutscene started then 7:00: if the kill event is already locked, keep `DreamBreak` (§4-1). No off-screen attacks (§4-3): telegraphs only on-screen. No elites during boss/hidden (§6-4).

Accessibility

Boss telegraphs multiplexed by color+shape+lead time (§4-3). Groggy/counter via sound+icon. Cut-in simplify toggle (§9-3). Hidden-boss time dual-shown with the top-left clock.

UX rationale

- **See at a glance:** in a fight a big boss bar appears top-center so "who I'm fighting" is clear — that's why the slot was kept empty. Time (top-left) and HP (top-left) stay put.
- **Operate / avoid mistakes:** the path to break the boss (crack→groggy) and the counter moment (0.7s) are shown separately, so "when to be greedy" is unambiguous.
- **Feel:** lucid counter → crack fills → groggy → harder hits: the "risk into reward" core amplified in the boss fight.
- **First vs repeat:** the trial banner spells out goal/time for newcomers; veterans hunt groggy to cut kill time.
- **Accessibility:** telegraphs by color+shape+time; cut-ins toggleable; time dual-shown.
- **Watch:** ① missing time because alarm is top-left → clock kept + hidden-boss time inset (dual); ② boss bar+crack+counter clutter → split top(HP)–below(crack)–near-player(counter); ③ effects hiding telegraphs → cut-in toggle, telegraph on top layer; ④ concurrent-end confusion → §4-1 priority, only a locked Dream Break keeps the cutscene.

Open questions

Boss HP segment count (currently 55% only). Counter window (A12) world-space vs HUD.

Acceptance

- ☐ On boss spawn the boss HP takes top-center while the top-left clock stays visible.
- ☐ In a hidden-boss fight the boss bar shows the inset time-left simultaneously.
- ☐ A successful boss-lucid counter increments crack; at 100 it flips to groggy.
- ☐ Colorblind mode distinguishes boss bar / crack / counter window by shape.

5.13 LVLUP — Level-up / Item pick · state: NormalRun (paused) overlay modal

Purpose

At level-up, **freeze combat** and pick one of 3 cards to grow the build — read a card in ~1s and decide **while seeing the current build/synergies**.

field	value
Enter	<code>RunTelemetry.level_reached</code> ++ (\$4-5)
Exit	card confirm → <code>NormalRun</code> / 7:00 → close then <code>DawnWake</code> (\$4-1)
Input context	paused
Priority	core

References

Source: *ui-ref harvest* — *interfaceingame* (boon/card pick).



What. Card grid, hover tooltip with effect, "Choose a Card" prompt, currency top. **Why.** Card = icon+name+effect reads in 1s, hover gives detail. →

applied: cards(4)·rarity(5)·delta/desc(6)·prompt(7).



What. Reward candidates as cards/icons with short effects. **Why.** Reward clarity. → applied: concise card effects.

Words: **pause** + **3–4 cards** (Vampire Survivors), **choice clarity** + **rarity color** (Hades), **review current build while choosing** (the 20MTD gap) → current-build panel (3).

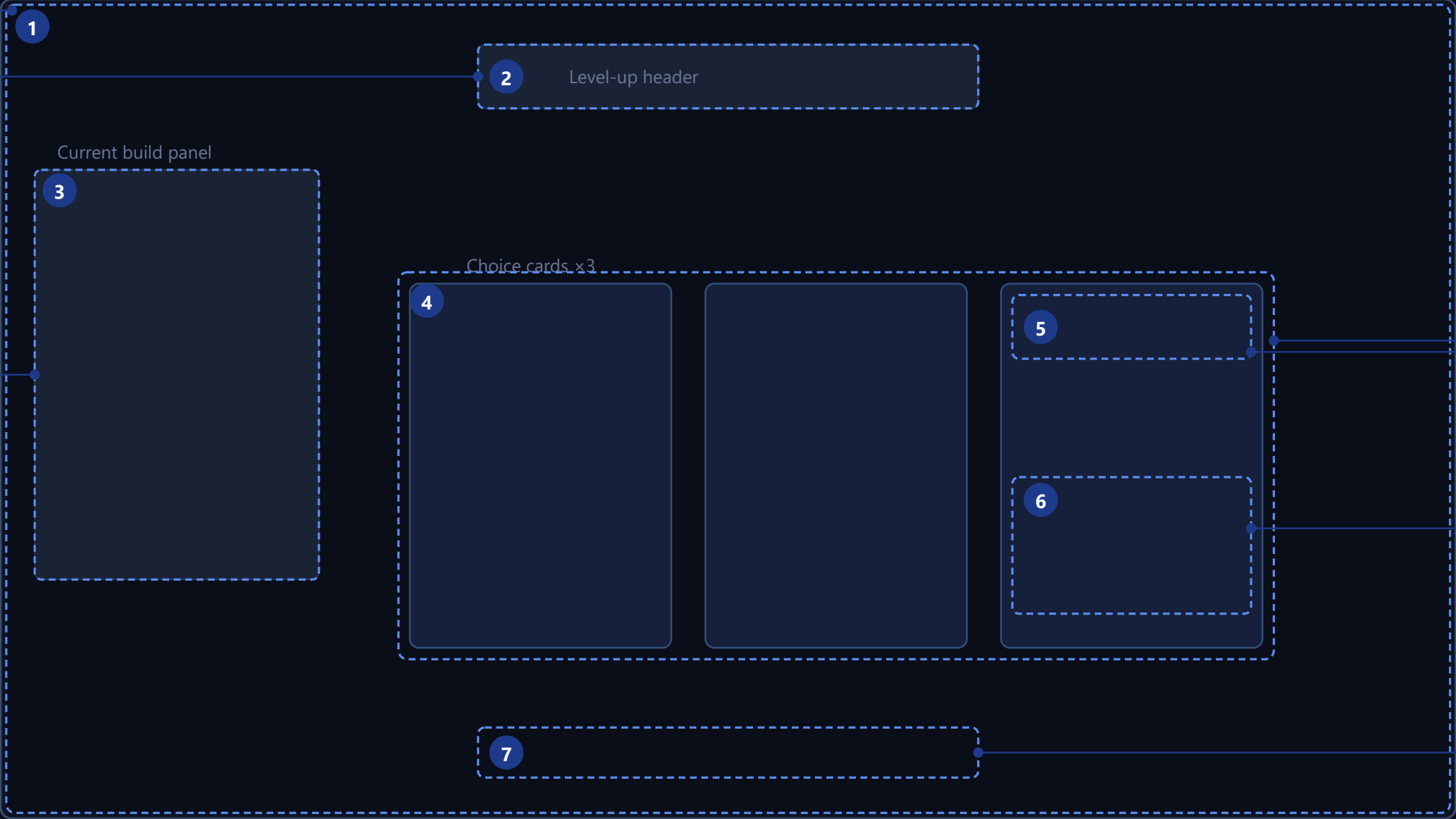
Wireframe

Level-up / Item pick (paused modal) — 16:9

1 **Pause dim**
combat frozen

2 **Level-up header**
top-mid · level n

3 **Current build**
owned6 · synergy



4 **Choice cards ×3**
center · 1s read

5 **Rarity-new/up**
color+shape+text

6 **Stat delta-desc**
+/- values

7 **Pick prompt**
bottom · click/A

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	B1	Pause dim	full	modal	combat freeze + dim	combat freeze (level-up)	freeze at once (0 input loss)	remove time pressure/strain	dim + input-block notice
2	B2	Level-up header	top-center	modal	"Level n reached" + flourish	<code>level_reached</code> (§12-2)	updates on level-up	what happened in one line	text+icon+sound
3	B3	Current build	left	modal	owned items (≤6)·level-synergy (read-only)	<code>Item.*</code> (§12-2), max 6(§4-5)	always visible before pick	choose by synergy	icon+name+level
4	B4	Choice cards ×3	center	modal	3 cards; icon+name+effect; focus emphasis; confirm	<code>RewardScreenRule.item_offer_count</code> (§12-2); 3 normal(§4-5)	1 card read ≤1s; pick ≤100ms	pick a direction fast	focus ring ≥3:1, labels
5	B5	Rarity-new/upgrade	on card	modal	rarity = color+border+corner; new/upgrade badge	<code>Item.rarity</code> (§12-2); odds(§4-5)	rarity by shape/text too	rarity & new-effect at once	color+shape+text
6	B6	Stat delta-desc	card bottom	modal	+/- numbers + level effect	<code>Item.effect</code> (§12-2); table(§4-5)	sign+value	what improves	sign+number
7	B7	Pick prompt	bottom	modal	device glyph + action	input(§2-5)	swaps on device change	no memorization	device glyph

State matrix (card B4)

element	default	hover/focus	selected	disabled/locked	loading	empty	error
Card (B4)	normal	emphasis+scale, ring ≥3:1	confirm + gem + short i-frame	Lucid card: lock+"lucid condition" pre-condition (§4-5)	dealing anim (≠ no-data)	"no choices" fallback	"gen error — retry"
Build panel (B3)	owned	item tooltip	N/A	owned 0 = "none yet"	enter fade	"no items"	"build load error"
Rarity (B5)	rarity color+shape	—	—	maxed item removed from pool (§4-5)	—	—	—

Input parity

action	mouse	key	pad	screen reader
focus card	hover	←/→	D-Pad/stick	"card 2/3, Moonlit Ribbon"
confirm	click	Enter	A/○	"Moonlit Ribbon, move +8%"
review build	hover	Tab	RB	"owned: Blue Pillow Lv.1"

No reroll (§4-5) — no reroll control.

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
B2	<code>level_reached</code> (§12-2)	—	<code>LevelReached</code>	"Level n"	last
B3	<code>Item.*</code> (§12-2), ≤6(§4-5)	—	<code>InventoryChanged</code>	icon+Lv	"none"

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
B4	<code>RewardScreenRule.item_offer_count</code> · <code>rarity_guarantee</code> (§12-2), <code>ItemPoolRule.*</code> (§12-2)	—	<code>LevelUpOffer</code>	3 cards	edge
B5	<code>Item.rarity</code> (§12-2), <code>ItemPoolRule.base_weight</code> (§12-2)	—	—	rarity+badge	Common
B6	<code>Item.effect</code> (§12-2), <code>table</code> (§4-5)	—	—	±value	"effect TBD"

Needs add: `LevelReached` , `InventoryChanged` , `LevelUpOffer` , `ItemPicked` (confirm; `stacking_rule` / `category_limit` per §12-2).

Navigation

Modal only. Confirm → `NormalRun` (I1). Multiple level-ups queue.

Edge cases

Pick rules (§4-5): first level-up = Common/Uncommon + ≥1 defense guaranteed / normal = ≥1 upgrade + ≥1 new / owned 6 = upgrades only / category-3 limit, maxed removed. Empty pool: 1 fallback reward `[TBD]` . 7:00 mid-pick (§4-1); close then `DawnWake` (no reward if unpicked). Rapid multi level-up: queue.

Accessibility

Rarity/new via color+border+corner icon+text (§9-3). Focus ring ≥3:1. Paused = no time pressure. Scalable text.

UX rationale

- **See at a glance:** freeze + dim focuses three cards; icon=type, color/border=rarity, short number=effect → reads in 1s.
- **Operate / avoid mistakes:** the left current-build lets you pick by synergy (the gap many survivor-likes miss); input glyphs at the bottom.
- **Feel:** confirm pops a gem + brief i-frames — the "I grew" beat; higher rarity = more flourish.
- **First vs repeat:** first level-up seeds defense so novices don't collapse; veterans build fast within the guarantee rules.
- **Accessibility:** rarity not color-only.
- **Watch:** ① picking without knowing synergy → build panel always on; ② colorblind rarity → color+border+icon+text; ③ 7:00 overlap loss → rule-based close then Dawn Wake; ④ maxed/category misclick → removed from pool / shown as upgrade.

Open questions

Empty-pool fallback type. Card-confirm i-frame length.

Acceptance

- ☐ Level-up freezes combat at once and shows 3 cards.
- ☐ owned<6 guarantees ≥1 new + ≥1 upgrade; owned=6 upgrades only.
- ☐ First level-up includes ≥1 defense item.

- ☐ Colorblind mode distinguishes rarity by shape/text.
- ☐ Pad-only focus & confirm.

5.I4 BOSSRWD — Boss reward (skill tree → 4-item pick) · state: BossReward (8s protect)

Purpose

Right after a boss/seal-trial kill, **grant the skill point**, then run ① **skill-tree node pick** → ② **a separate 4-item pick**. GDD §12-3: SP grant and item-pick UI are **separated**; skill nodes are not mixed into the 4-pick.

field	value
Enter	<code>BossDefeated</code> (§12-1) → SP +1 (§7-1)
Exit	step-2 (4-pick) confirm → <code>NormalRun</code> (I1)
Input context	8s protect (§7-1)
Priority	core

References

Source: *ui-ref harvest* — *interfaceingame* (skill tree / reward).



What. Upgrade rows/nodes, currency top-right, selected-item desc bottom-right. **Why.** Nodes+desc confirm effect/cost/prereq. → applied: node graph(4)·node state(5)·node info(6)·skill point(2).

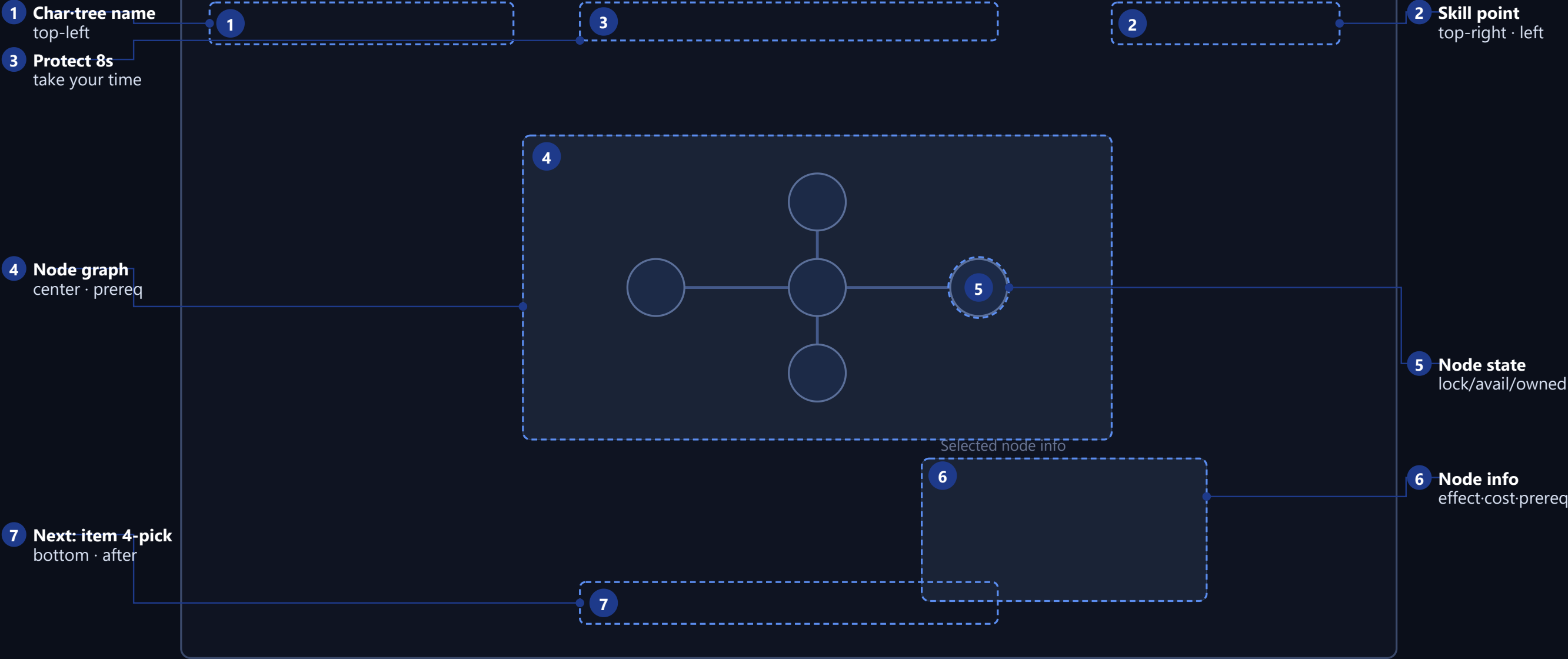


What. Reward items as a card/list to pick. **Why.** Reuse the same card grammar for the 4-item pick. → applied: 4-pick (C-CARD, 4 cards, Rare guaranteed).

Words: separate permanent growth (tree) from run reward (items) **by step** (\$12-3); no nodes in the 4-pick.

Wireframe (step 1: skill tree)

Boss reward step 1 — Skill tree (spend skill point) — 16:9



Step-2 4-item pick reuses §13 cards (C-CARD) as **4 cards, Rare guaranteed** (table below).

Legend (step 1 skill tree)

#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	C1	Char-tree name	top-left	always	the character's dedicated tree	<code>Character.id · SkillNode.character_scope</code> (§12-2)	shown on enter	whose tree	text+portrait
2	C2	Skill point	top-right	always	remaining SP; decrements on use	<code>RewardScreenRule.skill_point_delta</code> (§12-2), boss +1 (§7-1)	+1 at once on kill	what I can spend	number+icon
3	C3	Protect 8s	top	enter 8s	"take your time" + remaining	<code>RewardScreenRule.timeout_rule</code> (§12-2), 8s (§7-1)	8.0s protect	unhurried choice	text+countdown+sound
4	C4	Node graph	center	always	nodes + prereq edges; 4 nodes per char (§5)	<code>SkillNode.id · prerequisite · offered_after_boss</code> (§12-2)	pickable after kill (§4-5)	growth path	nodes+edges
5	C5	Node state	graph	always	locked/available/owned	<code>SkillNode.prerequisite · cost</code> (§12-2)	"available" when prereq+SP met	what can I take	lock/border/fill
6	C6	Node info	low-right	on focus	effect-cost-prereq	<code>SkillNode.effect · cost · prerequisite</code> (§12-2)	updates on focus	confirm before take	title+detail
7	C7	Next: 4-item pick	bottom	after SP use/skip	go to step 2	<code>RewardScreenRule.step_order</code> (§12-2/§12-3)	tree → 4-pick transition	reward order clear	button+prompt

Step 2: 4-item pick (reuse C-CARD)

After the tree, a **separate screen** for the 4-pick (§7-1, §12-3). Card component/states/a11y as I3 B4–B6, except: | item | value (GDD §) | | --- | --- | | cards | **4** (`RewardScreenRule.item_offer_count` , §4-5) | | guarantee | **≥1 Rare**, no skill nodes (§4-5, §12-3) | | Dream Break route | Blue Second-Hand · Starlight Lens · lucid-line weighting (§4-5) | | elite reward (ref) | 3 cards, ≥1 Uncommon (§4-5) |

State matrix (node C5 / 4-pick card)

element	default	hover/focus	selected	disabled/locked	loading	empty	error
Node (C5)	owned=fill/else=outline	emphasis+info	just-taken	prereq unmet=lock+"need prereq" / low SP="point short"	enter fade	N/A: 4 nodes	"tree load error"
4-pick card	normal	emphasis+ring	confirm+gem	maxed hidden (§4-5)	dealing anim	"no choices"	"gen error"

Input parity

action	mouse	key	pad	screen reader
focus node	hover	arrows	D-Pad/stick	"Steady Counter, cost 1"
confirm node	click	Enter	A/○	"Steady Counter taken, SP 0"
next/skip	click	Esc/Enter	B/A	"to 4-item pick"
pick item	click	Enter	A	"Starlight Lens selected"

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
C2	<code>RewardScreenRule.skill_point_delta</code> (§12-2)	<code>BossDefeated</code> (§12-1)	<code>SkillPointChanged</code>	int	0

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
C4-C5	SkillNode.* (§12-2)	BossDefeated (§12-1)	SkillTreeOpened , NodeStateChanged	graph	read-only
C6	SkillNode.effect · cost · prerequisite (§12-2)	—	NodeFocused	detail	"select node"
C7/4-pick	RewardScreenRule.step_order · item_offer_count · rarity_guarantee (§12-2)	BossDefeated (§12-1)	RewardStepAdvanced , BossItemOffer	4 cards	edge

Needs add: SkillPointChanged/SkillTreeOpened/NodeStateChanged/NodeFocused/NodePicked/RewardStepAdvanced/BossItemOffer/ItemPicked .

Navigation

step 1 (tree) → C7 → step 2 (4-pick) → confirm → NormalRun (I1). Seal-trial rewards reuse this in shortened form (§7-2: Seal 3 Trial = Rare pick).

Edge cases

Reward order fixed (§7-1): on HP0, BossDefeated grants SP+1 & purify bonus first → during 8s protect, tree → then 4-pick. SP 0/skip: proceed without taking (SP carries). 8s end: auto-advance, unspent SP kept (assumed) [TBD] . Dream Break route weighting (§4-5).

Accessibility

Node state via color+lock/border/fill (§9-3). 8s protect = no time pressure. Info panel scalable; focus order numbered; 4-pick inherits I3 a11y.

UX rationale

- **See at a glance:** separate permanent growth (tree) from run reward (4-pick) by step so they don't blur. The tree draws nodes+edges to show "what I can take now / what opens next".
- **Operate / avoid mistakes:** focusing a node shows effect/cost/prereq before confirm; the 8s protect cuts misclicks and unspent points carry.
- **Feel:** granting the skill point up front cements "I beat the boss".
- **First vs repeat:** protect lets novices read; veterans take recommended nodes (§7-1) fast then go to the 4-pick.
- **Accessibility:** node state not color-only.
- **Watch:** ① node/item confusion → step separation, no nodes in the 4-pick; ② prereq/point misclick → color+lock+text; ③ reward-order tangle → SP first, order enforced; ④ 8s feeling rushed → "take your time" + points kept.

Open questions

Unspent SP/items at 8s end. Per-character tree as separate screen vs shared frame (currently shared).

Acceptance

- ☐ On kill, SP+1 is granted first, then tree → 4-pick in order.
- ☐ The 4-pick is 4 cards with ≥1 Rare and no skill nodes.
- ☐ Prereq-unmet/point-short nodes show color+lock+text.
- ☐ During 8s protect you can choose slowly; skipping keeps points.

- ☐ Pad-only node & 4-pick selection.

5.15 PAUSE — Pause menu · state: NormalRun / BossFight paused

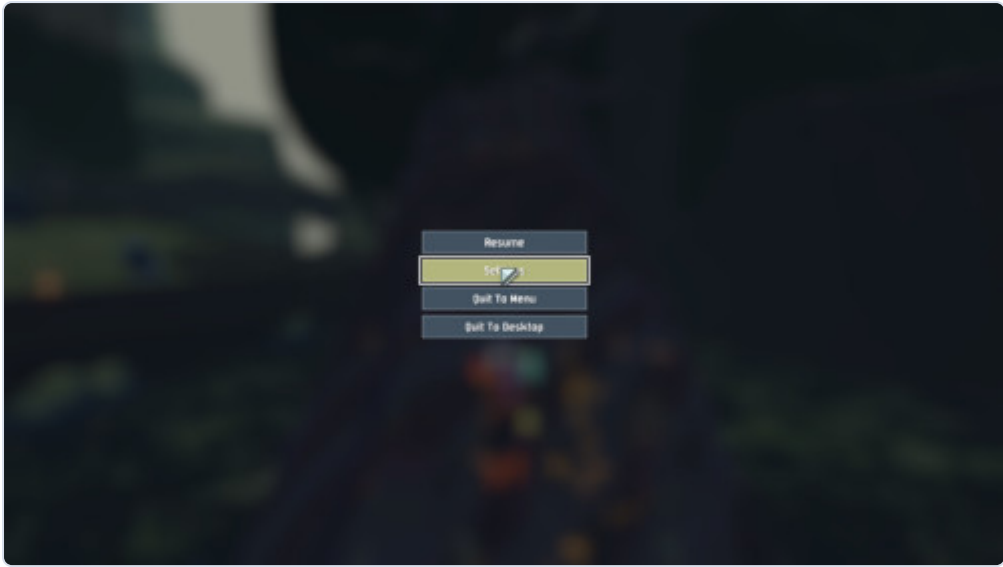
Purpose

Mid-combat ESC/Start freezes the field; choose resume/settings/controls/main/quit. Show a run summary to re-orient.

field	value
Enter	ESC / Start (in combat)
Exit	resume → combat / settings → O4 / main-quit → after confirm
Input context	paused
Priority	core

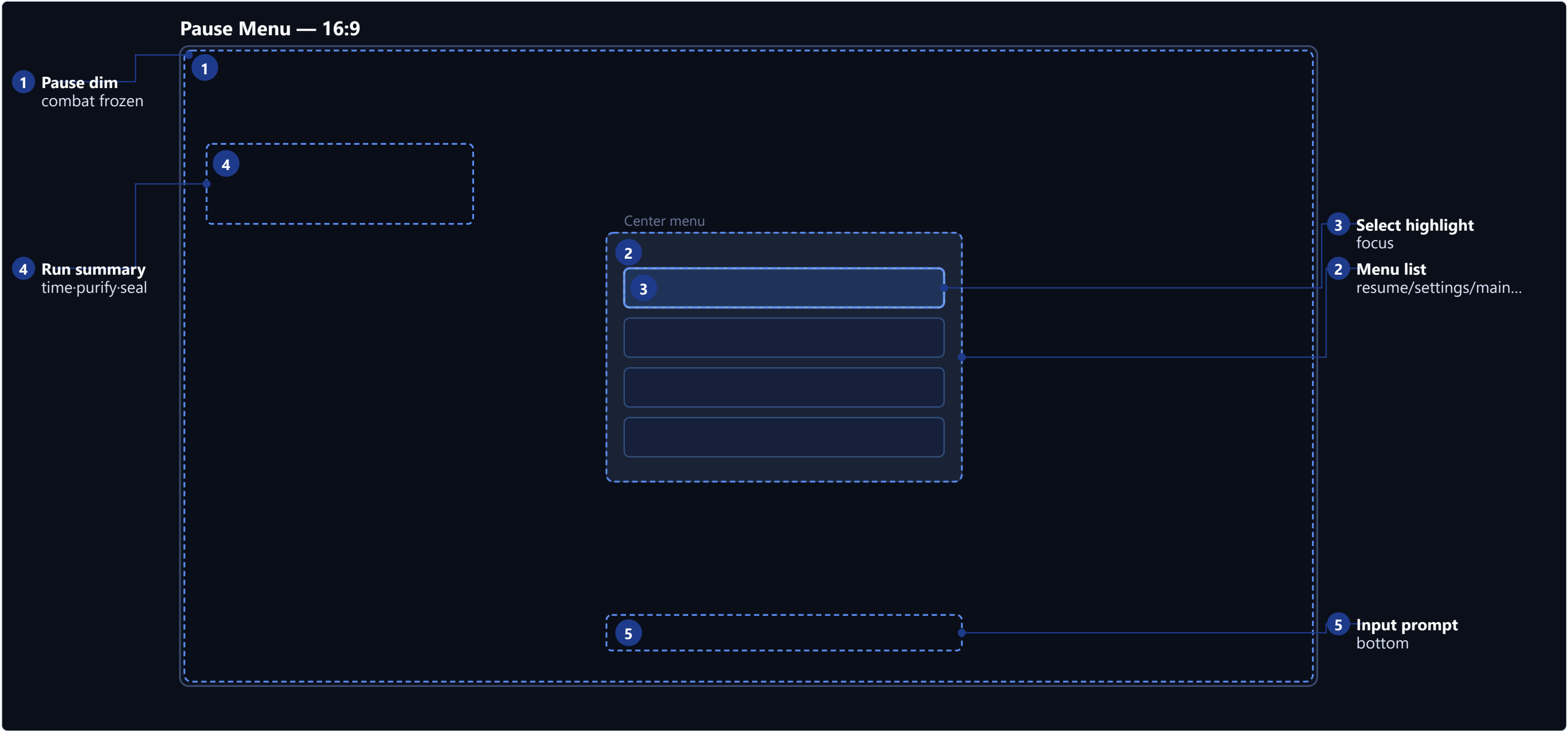
References

Source: *ui-ref harvest — interfaceingame (pause)*.



What. A dimmed field with a center vertical menu (Resume/Settings/Quit to Menu/Quit to Desktop). **Why.** Pause is best as a short center vertical menu — resume fast or leave. → applied: dim(1)-center menu(2)-highlight(3)-input(5).

Words: add a **run summary(4)** (time/purify/seal) to re-orient on resume; main/quit lose progress → confirm.



#	code	element	position	shown	behavior/state	data binding	criterion	UX intent	a11y
1	Z1	Pause dim	full	menu	combat freeze + dim	freeze (ESC)	freeze at once	clearly stopped	dim + input-block notice
2	Z2	Menu list	center	menu	resume/settings/controls/to main/quit	menu (static)	select ≤100ms	fast resume/leave	label+focus
3	Z3	Select highlight	item	menu	focus emphasis + SFX	input focus	emphasis on move	where am I	border+color+sound
4	Z4	Run summary	top-left	menu	time-left-purify-seal progress	<code>run.timer</code> (§4-1)·purify(§4-2)·seal(§4-2)	snapshot at open	re-orient	text+numbers
5	Z5	Input prompt	bottom	menu	device prompt	input(§2-5)	swaps on device	how to operate	device glyph

State matrix (Z2/Z3)

element	default	hover/focus	pressed	disabled	loading	error
Menu item (Z2)	normal	emphasis+SFX, ring ≥3:1	pressed	—	N/A	"menu error"
Highlight (Z3)	Resume default focus	tracks	—	—	—	—
Run summary (Z4)	values	—	—	—	snapshot at open	missing = "—"

Input parity

action	mouse	key	pad	screen reader
open/close	—	ESC	Start	"paused"
move	hover	↑ / ↓	D-Pad	"Resume, 1/5"
run	click	Enter	A/○	"Resume activate"
main/quit	click	—	—	"To main, confirm progress loss"

Data binding

code	field (GDD §)	event	UI-proposed (needs add)	format	fallback
Z2	menu (static)	—	<code>PauseFocusChanged</code>	items	default
Z4	<code>run.timer</code> (§4-1)·purify(§4-2)·seal(§4-2)	<code>PurgeGained</code> (§12-1)	<code>RunSummarySnapshot</code>	numbers	"—"

Needs add: `PauseOpened/PauseClosed` , `RunSummarySnapshot` .

Navigation

Resume → combat. Settings → O4 (re-paused on return). To main/quit → confirm modal then leave (no I6, progress forfeit). Controls → control overlay.

Edge cases

Main/quit forfeit progress (irreversible) → confirm. Pause allowed in boss fight (solo); co-op (out of scope) differs. 7:00 arriving while paused: on unpause, apply §4-1 priority.

Accessibility

Focus = color+border+sound; main/quit confirm states the outcome in words; summary shows numbers; subtitles/text scale.

UX rationale

- **See at a glance:** a short center vertical menu, Resume on top and default-focused; the left run summary re-orients "how far am I".
- **Operate / avoid mistakes:** to-main/quit forfeit progress, so they confirm.
- **Feel:** clarity over flourish; resume fast.
- **First vs repeat:** a controls entry lets novices check inputs anytime.
- **Accessibility:** outcome (loss) stated in words.
- **Watch:** ① accidental quit losing progress → confirm modal; ② default focus on a dangerous item → default Resume; ③ 7:00 handling confusion → §4-1 priority on unpause.

Open questions

Showing 7:00 arrival while paused (instant result vs on-unpause). Controls overlay vs separate screen.

Acceptance

- ☐ ESC/Start freezes the field and shows the center menu.
- ☐ Default focus is Resume.
- ☐ To-main/quit show a progress-loss confirm.
- ☐ Run summary (time/purify/seal) is shown.

5.I6 RESULT — Result / Summary (3 branches) · state: DawnWake · DreamBreak · FailedWake

Purpose

Close the run with the **emotion matching how it ended**, show **final build and run stats separately**, then meta rewards/unlocks. Even on failure, show the next goal.

field	value
Enter	AlarmReached →DawnWake / DreamBreakAchieved →DreamBreak / HP0→FailedWake (§4-1)

field	value
Exit	Retry→O2 / Main / Leaderboard(O6)
Input context	non-combat, focus
Priority	core

References

Source: ui-ref harvest — interfaceingame (results).



What. "Defeat!" title, left Stats panel (time/kills/damage...), right Info (class/killed-by/Items Collected/Unlocked), Continue. **Why.** Separate build (items/unlocks) from stats (records) so neither cramps. → applied: ending badge(1)·final build(3)·run stats(4)·currency/unlock(6).



What. A failure screen with reached records. **Why.** Even failure closes with a reached point / next goal. → applied: FailedWake branch (2) + next goal.

Words: **results must partition build vs stats** (Brotato). Dream Break = triumphant, Dawn Wake = relief, Failed = calm — same screen, different emotion.

Wireframe

Result / Summary (Dawn Wake · Dream Break · Failed Wake) — 16:9

1 Ending badge
3 branches

1 Result title / ending badge

2 Ending pose
per-branch

2 Character ending pose

6 Currency-unlock
shards/stardust/core

6

3 Final build (partition A)

4 Run stats (partition B)

5

7

3 Final build
weapon-items-level

4 Run stats
time-kills-evades

5 Records-splits
DB time-boss

7 Action buttons
retry/main/ranks

Legend

#	code	element	position	shown	behavior/state (per branch)	data binding	criterion	UX intent	a11y
1	D1	Ending badge	top-center	always	DawnWake/DreamBreak/FailedWake — text+color+icon	branch(\$4-1), <code>AlarmReached</code> · <code>DreamBreakAchieved</code> (§12-1)	branch badge on enter	how it ended	text label required
2	D2	Ending pose	left	always	per-branch pose+line (§9-4)	<code>Character.id</code> (§12-2), lines (§9-4)	branch pose/line sync	character close	subtitle+voice, cut-in toggle
3	D3	Final build (A)	top-right	always	weapon level + items(≤6) + level	<code>CombatWeapon</code> · <code>Item</code> · <code>weapon_level</code> · <code>level_reached</code> (§12-2)	accurate at end	this run's build	icon+name+level
4	D4	Run stats (B)	mid-right	always	time-kills-purify-lucid evades-boss time	<code>RunTelemetry.*</code> (§12-2)	telemetry accurate	performance numbers	line-aligned+numbers
5	D5	Records-splits	low-right	by branch	DB Time-boss splits-hidden unlock-no-damage-highest purify (§10)	<code>dream_break_stage_times</code> (§12-2), records (§10)	achieved only, else "—"	competition/PB	text+number+PB badge
6	D6	Currency-unlock	low-left	always	dream shards/stardust/lucid core + unlocks (§2-3)	meta currency (§2-3)	shown at once on end	repeat motivation	icon+number+unlock text
7	D7	Action buttons	low-right	always	Retry/Main/Leaderboard	routing	≤100ms, default focus	next action	button+prompt

Per-branch differences

branch	D1 badge/color	D2 line (§9-4)	D5 emphasis	special
<code>DawnWake</code>	dawn color, "Dawn Wake"	"...it was a dream. <i>phew</i> ."	survival-purify	7:00 + boss kill = no boss reward (§4-1)
<code>DreamBreak</code>	whiteout, "Dream Break"	"wait, I really thought I shot it..."	DB Time-splits-no-damage	true-ending emphasis, lucid core
<code>FailedWake</code>	cool color, "Failed Wake"	cold sweat	reached point-next goal	failure motivation (§13-3 #8)

State matrix (D7 / D5)

element	default	hover/focus	selected	disabled	loading	empty	error
Action button (D7)	normal	emphasis+ring	default focus (Retry)	leaderboard offline = dim+"Offline"	enter fade	N/A: 3 buttons	"transition fail — retry"
Records (D5)	values	item detail	new record = badge	not achieved = "—" (branch n/a)	"calculating"	"no records"	"record load error"
Currency (D6)	shown	—	new unlock = emphasis	—	settle anim	"none"	"settle error"

Input parity

action	mouse	key	pad	screen reader
focus button	hover	Tab/arrows	D-Pad	"Retry (default)"
run	click	Enter	A/○	"Retry activate"
record detail	hover	arrows	RB	"Dream Break Time 9:12, new record"
main	click	Esc	B	"Main menu"

Data binding

code	field (GDD \$)	event	UI-proposed (needs add)	format	fallback
D1	branch(\$4-1)	AlarmReached · DreamBreakAchieved (\$12-1)	RunEnded{result}	badge	"result TBD"
D2	Character.id (\$12-2), lines(\$9-4)	(above)	—	pose+subtitle	default
D3	CombatWeapon · Item · weapon_level · level_reached (\$12-2)	—	RunBuildSnapshot	icon+Lv	"none"
D4	RunTelemetry.* (\$12-2)	—	RunStatsSnapshot	numbers	0
D5	dream_break_stage_times (\$12-2), records(\$10)	DreamBreakAchieved (\$12-1)	RecordSplits	time/number	"—"
D6	meta currency(\$2-3)	—	RewardsGranted , UnlocksGranted	number+unlock	0

Needs add: RunEnded{result}/RunBuildSnapshot/RunStatsSnapshot/RecordSplits/RewardsGranted/UnlocksGranted .

Navigation

Retry→O2 / Main / Leaderboard(O6, matching filter). Default focus **Retry**. Pad B/Esc = Main.

Edge cases

Concurrent ends (\$4-1): 7:00 + kill → DawnWake priority. Dream Break cutscene entered then 7:00 → keep DreamBreak. Offline: leaderboard button dim+reason, records cached. New record: D5 badge+sound. FailedWake: still grants some reward (\$2-3) + next goal.

Accessibility

Ending branch via text label + icon, not color only (\$9-3). Cut-in/ending toggle. Subtitle size/bg. Unachieved records as "—" (no blanks). Default focus/order clear.

UX rationale

- **See at a glance:** a big top badge conveys "how it ended" with emotion; below, this-run's build (weapon/items) and play record (time/kills/evades) split left/right so neither cramps.
- **Operate / avoid mistakes:** three buttons, default focus on Retry for one-tap replay; leaderboard unavailable = dim+reason.
- **Feel:** Dream Break = triumphant whiteout + true-ending line; Dawn Wake = relief; failure = calm — per-branch emotion. New record = badge.
- **First vs repeat:** failure shows reached point + next goal so you want another go; veterans chase DB Time/splits/no-damage.
- **Accessibility:** branch stated in text; effects toggleable.
- **Watch:** ① build/stats blur → partition (Brotato); ② failure feeling like a dead end → reached/next-goal/partial reward (\$13-3 #8); ③ colorblind branch → badge text; ④ concurrent-end confusion → \$4-1 priority.

Open questions

FailedWake meta reward ratio. Leaderboard axis exposure.

Acceptance

- ☐ Each branch shows the right badge/pose/line.
- ☐ Final build and run stats are shown in separate areas.
- ☐ Dream Break shows DB Time-splits:no-damage records.
- ☐ Colorblind mode distinguishes the ending branch by text/icon.
- ☐ Default focus is Retry; pad-only restart works.

6. Cross-cutting rules

- **Navigation model**
- Out-game (O1–O6): focus-driven. Always set a default focus (Main=Single, Character=Kohaku, Stage=LD-001/Normal, Result=Retry, Pause=Resume). Back/cancel = Esc/B. Irreversible (quit/to-main/unlock/run-start loss) confirms.
- In-game (I1–I2): real-time. Only modals (level-up I3) / protection (reward I4) / pause (I5) overlay; combat freezes during them.
- Switching input device swaps every C-PROMPT glyph at once (no memorization).
- **Global empty / loading / error:** **loading** distinguishes **enter-anim** from **no-data** ("loading.../syncing..." vs fade-in). **error** states **problem + fix** ("record load error — retry"); missing value = last value + warning border. **empty** is never blank — give a reason ("—", "undiscovered").
- **Accessibility (GDD §9-3):** colorblind palette (enemy/ally bullets, hazard zones by shape+color), shake/flash/post-processing strength, hit-judgment display, ult cut-in simplify, auto-aim bias, lucid-evade assist, subtitles (size/bg/speed). **No screen conveys state by color alone.** Settings (O4) is the gateway to all of it.
- **BM consistency (GDD §10):** no paid power / no gacha → **no cash-purchase UI on any screen.** Meta (O5) unlocks via play-earned currency only.
- **Layout / safe area:** 16:9 (1920×1080). HUD at edges/top/bottom, center clear. Top-left (status) / bottom (resources) / top-center (boss) slots have fixed meaning across screens (P4). 21:9-4:3 scale within safe area (later).

6-1. UX synthesis (GDD §13-3 risk linkage)

UX risk (GDD §13-3)	screens	decision / mitigation	verification
danger unseen in volume (#5)	I1, I2	HUD to edges, center clear; hazards color+shape+sound	180-enemy + colorblind playtest
time awareness vs top-left diegetic timer	I1, I2	alarm fixed top-left + final-60s edge/sound + hidden-boss time dual	1s read from any screenshot
late-game tedium (#3)	O3, I1, I6, O5	purify/seal visualize early boss spawn; Dream Break goal in stage/result; meta long-term goals	progress awareness · retry rate
lucid evade too hard/spammy (#4)	I1, I2	combo + near/lucid reward tiers; counter window visible	novice accidental · veteran intentional
Dream Break too hard (#8)	I2, I6	seal progress · hidden-boss approach; failure → next goal	reach rate · retry rate
survivor-like clone look (#2)	all	lucid evade · alarm clock · Dream Break · per-character tree front-and-center	differentiation survey
scope creep (#7)	O2, O3	VS Kohaku/Toko/LD-001 active, rest locked	out-of-scope unbuilt check

UX risk (GDD §13-3)	screens	decision / mitigation	verification
monetization confusion (BM §10)	O5, all	no cash UI, currency unlocks only	no purchase screen
license/official confusion (#1)	O1, O2	character-name/display license area — layout only, legal separate	display-guideline review
co-op time-stop disturbing others (#6)	(out of scope)	unbuilt in VS; per-player/team/joint effect split at co-op build	— (out of UX scope)

1 and #6 are partly outside UX (legal / co-op build) and flagged as such. The rest are handled by this spec's layout/state/feedback rules.

7. Open questions (aggregate)

#	screen	question
Q1	I1	low-HP warning threshold (% — currently 25%)
Q2	I1	promote XP bar to a §9-2 HUD element?
Q3	I1	purify(A7) vs seal(A3) meaning overlap
Q4	I2	boss HP phase segment count / counter-window placement
Q5	I3	empty-pool fallback reward / card i-frame length
Q6	I4	unspent SP/items at 8s protect end
Q7	I6	FailedWake meta reward ratio / leaderboard axis exposure
Q8	O2/O3	unlock-condition text / stats radar vs bars / difficulty lock
Q9	O4	accessibility presets / live-apply option scope
Q10	O5	codex spoiler policy / permanent tree vs run tree visual split
Q11	O6	co-op record exposure / cheat policy / season reset
Q12	global	21:9-4:3 safe area · HUD scale
Q13	global	data contract: add §5 "UI-proposed" events / runtime state to GDD §12-1/§12-2

Full list of needed runtime fields/events in the decision tracker (`lucid_dawn_ui_ux_decisions.md` §4). No cash-payment events (§10).

8. Version history

version	date	change
v1.0 (EN master)	2026-06-30	Full out-game(O1–O6)+in-game(I1–I6) annotated-wireframe spec, web-harvested refs (interfaceingame, 8 games), design tokens + decision tracker, engine binding (App. D), usability test plan (App. E). 13 wireframes lint+render pass. Korean/Chinese versions derived from this master.

Appendix A. Methodology (rationale)

The UX rationale sections reason with the frameworks below but never name them in the body (plain-language rule). | framework | used for | | --- | --- | | Celia Hodent, *The Gamer's Brain* | spine of every UX rationale (feedback/clarity/form-follows-function/consistency/error prevention + motivation/emotion/flow) | | Nielsen's 10 heuristics | status visibility, consistency, error messages (problem+fix), recognition over recall | | Pinelle game usability heuristics (CHI 2008) | camera/control/hidden-status/micromanagement failure modes | | Desurvire PLAY / HEP | fun/immersion lens | | Fagerholt & Lorentzon, *Beyond the HUD* | HUD element classification (diegetic alarm clock vs meta/non-diegetic bars) | | Swink *Game Feel*; Jonasson & Purho "Juice it or Lose it" | feel stack + juice budget + readability under chaos | | Xbox XAG; IGDA GASIG Top Ten; Game Accessibility Guidelines | color-independence, subtitles, shake/flash toggles, remap, difficulty/speed |

Appendix B. Reference index / harvest recipe

- Harvested with **Game UI Reference CLI (`ui-ref`)** from **interfaceingame.com** per-game pages: Hades · Risk of Rain 2 · Honkai: Star Rail · Slay the Spire · Returnal · Hollow Knight · Moonlighter · Destiny 2. Curated by screen type into `references/ui/web-refs/<category>/`. Personal research citations, not a redistributable asset pack. Manifest: `ui_research/manifests/local_ui_refs_manifest.md`.

```
# one-time: pip install playwright && playwright install chromium
# put interfaceingame per-game URLs in ui_research/urls.txt, then
ui-ref collect --browser --site interfaceingame --scroll 8 --download-gallery-assets --download-asset-limit 10
# gap-fill specific screens:
ui-ref collect --browser --site interfaceingame --download-title-contains defeat --download-title-contains paused --download-title-contains audio
ui-ref scan-local
```

Game UI Database is SPA + Cloudflare → headless harvest blocked (both the `scrn=` filter and `gameData.php?id=` per-game time out). We used interfaceingame per-game; direct survivor-likes are described in words in §B-2.

B-2. Direct survivor-like patterns (words — not captured)

The harvested games are adjacent roguelites, so direct **bullet-heaven / survivor-like** conventions are noted here. (Capture later from GUIDB Soulstone Survivors = `gameData.php?id=2403` , etc., subject to the headless limit above.) | screen | direct title | pattern to adopt / contrast | our callout | | --- | --- | --- | --- | | HUD | Vampire Survivors | **timer top-center, XP = top full-width strip**, HP near player, ability icons in one corner, minimal chrome | I1: XP A1 adopted; alarm top-left (deliberate exception) | | HUD | 20 Minutes Till Dawn | **manual aim** (not auto-target) → aim feedback/reticle, active abilities | I1: auto-aim bias option (§9-3) + skill slots A8 | | Level-up | Vampire Survivors / Brotato | **pause + 3–4 cards**, rarity color, instant read, brief i-frame on confirm | I3: paused cards B4, rarity B5 | | Level-up | 20 Minutes Till Dawn | **review current build/synergy while choosing** (the gap modders fix) | I3: current-build panel B3 | | Result | Brotato | **partition build vs stats**, meta currency/unlock surfaced | I6: D3 build / D4 stats split | | Inventory/build | Soulstone Survivors / Halls of Torment | many stacked items **scale-down + stack count**, rarity color, full build panel on pause | I3 B3, I6 D3 | | Boss | LoL Swarm (survivor mode) | top horizontal boss bar **takes the timer slot**, phase segments | I2: boss HP A10 (ours takes the empty top-center slot) |

Deliberate differentiation (avoid clone look, §13-3 #2): lucid evade (counter window A12) · alarm clock (diegetic top-left) · Dream Break (hidden boss + time dual A13) · per-character skill tree (I4). We foreground what the direct titles don't.

Appendix C. Build (shareables) + wireframe validation

```
pip install markdown python-docx
python <skill>/templates/build_pdf.py lucid_dawn_ui_ux_spec.en.md --css <skill>/templates/spec-pdf.css
python <skill>/templates/build_docx.py lucid_dawn_ui_ux_spec.en.md # render wireframes/*.png first
```

All 13 wireframes pass `validate_wireframe.py` (XML, background rect, ≤10 callouts, badges twice, leaders ≤4 pts non-crossing, gutter ≤6/side, no off-canvas) + headless-Chrome render eyeball. flow.svg is a wireflow (exempt from the callout linter).

Appendix D. Implementation binding — Unity 6 UI Toolkit × DOTS/ECS

Implements §5 data binding **event-driven, no-GC** on the chosen stack (Unity 6 · DOTS · UI Toolkit). Authoritative refs: `unity-dots-manual` , bullet-hell rendering recipe. Final decisions = TDD.

- **Roles:** ECS (sim, jobs/Burst) is the single source of runtime state (`RunState` , `PlayerRuntime` , boss/trial runtime — §7 add). UI Toolkit (managed, main thread) renders. A **bridge** `SystemBase` (main thread) reads ECS and pushes to UI **only on change** — the real implementation of "event-driven".
- **No-GC / no per-frame rebuild** (HUD must not cost at 180 enemies, P3): state as `IComponentData` singletons; bridge checks ECS **ChangeVersion** and updates only changed labels; cache `VisualElement` refs (no per-frame `Q<>()`); no LINQ/boxing in the loop; timer text updates per **second**, not per frame.
- **World vs screen space:** fixed HUD = screen overlay (UIDocument/PanelSettings); player/enemy-anchored (lucid combo A6, counter window A12, **damage numbers**) = world anchors — **heavy world text (damage numbers) goes through the project's mass render path, not UI Toolkit**; UI Toolkit only for the few HUD-anchored ones.
- **Events ↔ ECS:** §5 "UI-proposed" events implemented as bridge change-detection (DOTS-friendly); GDD §12-1 events (`BossDefeated` , etc.) received as ECS events/singleton flags driving UI transitions. Modals/reward = sim pause synced (system-group stop).
- **Multi-res / input / localization:** PanelSettings scale mode + safe area (tracker D-18); gamepad via UI Toolkit focus navigation (ring ≥3:1) + Input System; Unity Localization string tables (KR/EN/CN length, subtitle ~≤38/line, locale number format).
- **Checklist:** [] 0 alloc / minimal reflow at high density · [] update only on change (profiled) · [] mass world text off UI framework · [] sim pause synced during modals · [] pad focus nav on every screen.

Appendix E. Usability test plan (core loop)

The UX rationale is heuristic reasoning; this is how it gets validated. Wireframe-only claims (readability-under-chaos) MUST be re-validated in an engine build.

- **Goal/hypotheses:** pass the core loop (menu→combat→level-up→boss→reward→result) without getting lost; key legibility: ① read time-to-7:00 ≤1s, ② identify hazard/player/pickups at 180 enemies, ③ read a card ≤1s, ④ understand reward order (SP→tree→4-pick).
- **Method/participants:** moderated think-aloud, 6–8 — survivor-like novice 3 + veteran 3 + colorblind 1–2 (separate session). Build: VS scope.
- **Task scenarios ↔ acceptance:** | # | task | success | links | | --- | --- | --- | --- | | T1 | from start to run begin | ≤3 min, unguided, pick char+stage→start | O1/O2/O3 | | T2 | (random pause) "how long until 7:00?" | correct ≤1s | I1 A2 (§3-2) | | T3 | pick one level-up card | read ≤1s, **observe build-panel use** | I3 B3/B4 | | T4 | attempt lucid evade | novice accidental + combo seen / veteran intentional | I1 A6 (§3-2) | | T5 | boss fight | boss bar seen, crack→groggy & counter **used** | I2 A10/A11/A12 | | T6 | boss reward | **understand SP→tree→4-pick order**, node/item distinct | I4 | | T7 | result | recognize branch, build vs stats split, Retry restarts | I6 |
- **Metrics:** per-task success/time/errors · SUS · fun/engagement survey · **readability-under-chaos** (frozen high-density screenshot id accuracy, separate colorblind session) · accessibility (colorblind id / toggle behavior).
- **Pass bar / iteration:** core tasks (T1·T2·T3·T6) success ≥85%, T2 1s-read ≥90%, colorblind lethal id =100%. Misses → adjust the corresponding number in the tracker and re-test; reflect in §8.
- **Limit:** this is the *test design*. A real clickable prototype / sessions / SUS tally need build + people — not performed here.